EART 22201 Igneous minerals and processes

Optics and mineralogy



Lesson 2

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Optics and mineralogy

1.1 Light and the polarizing microscope

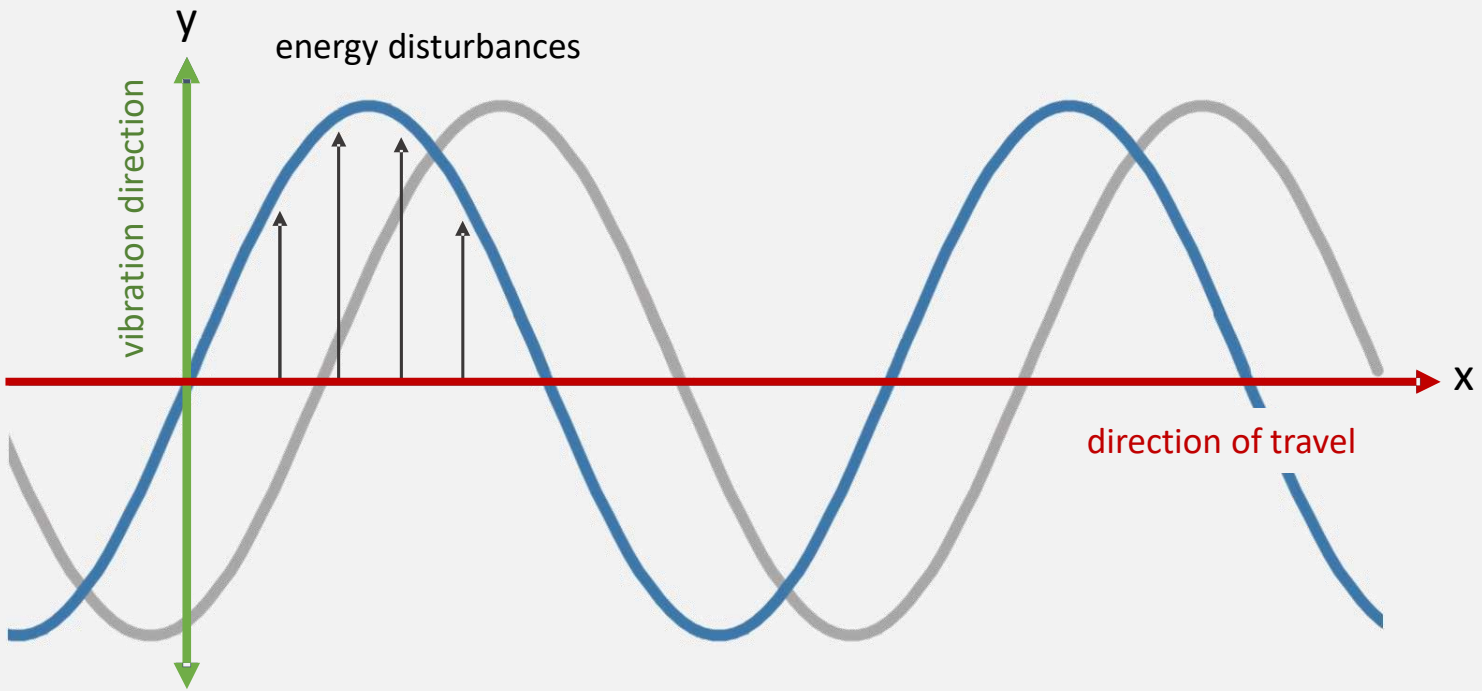
1.2 The refractive index

1.3 Relief in minerals



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Light is an electromagnetic wave

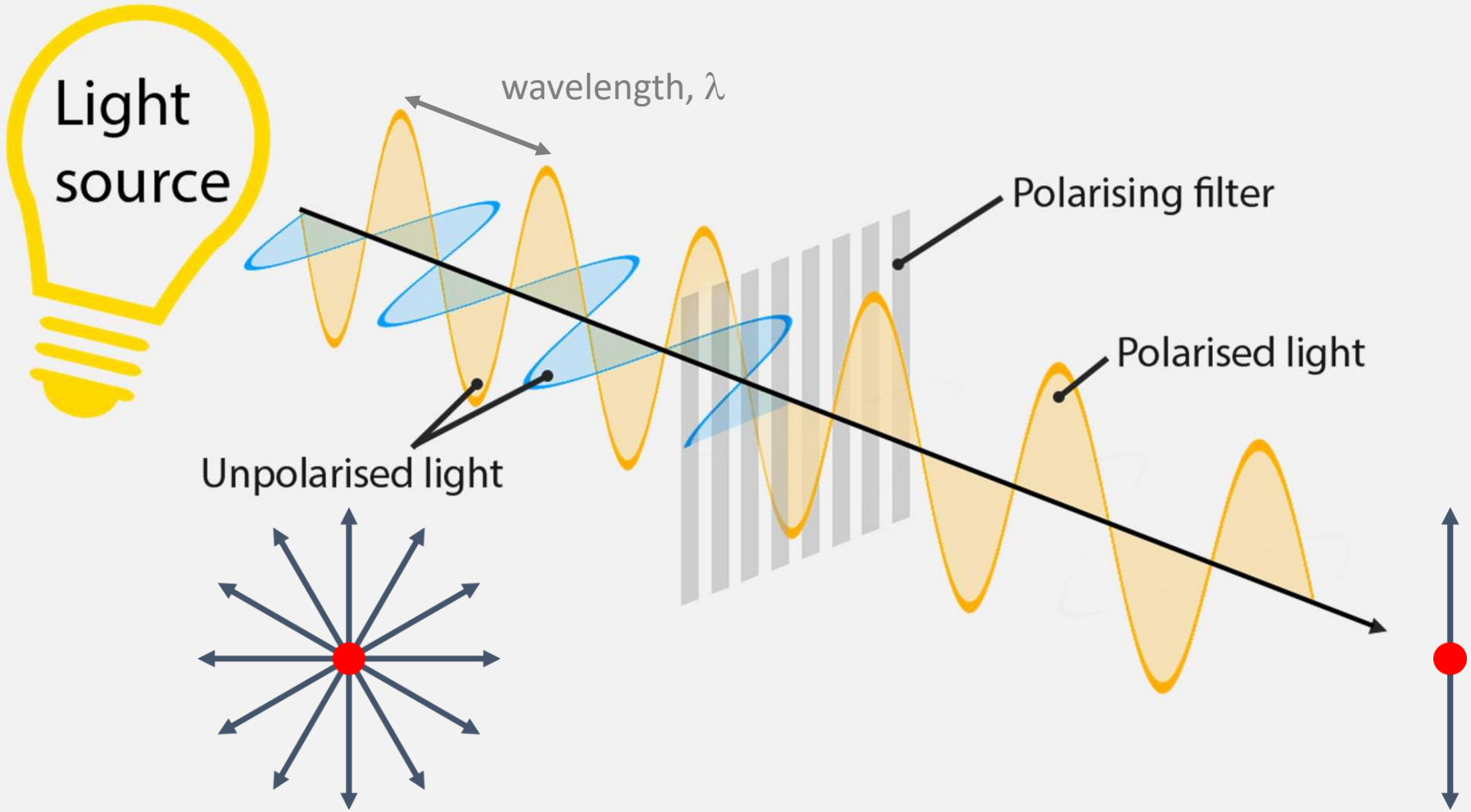


In minerals, energy disturbances cause vibrations.

The vibration direction is always perpendicular to the direction of travel.

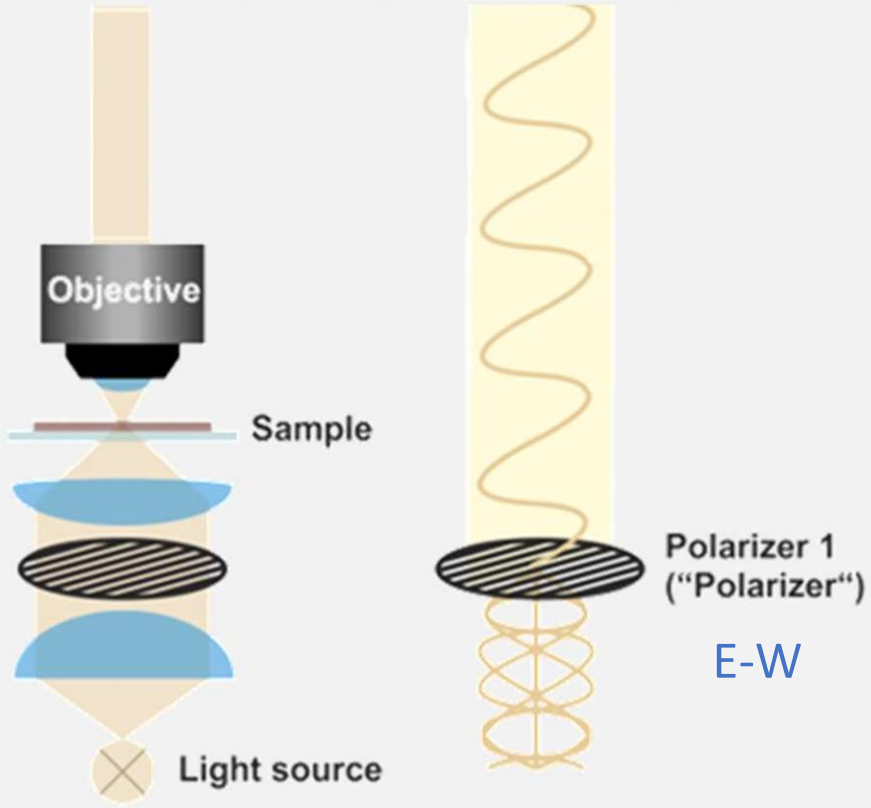
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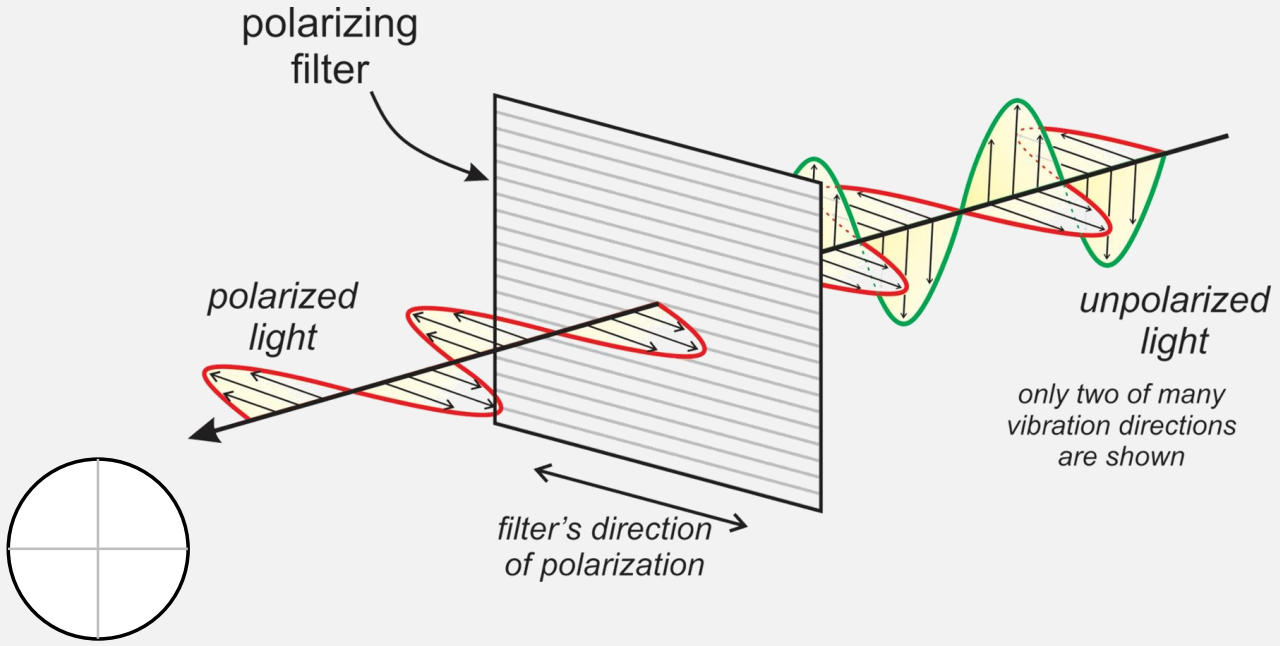
Polarized light



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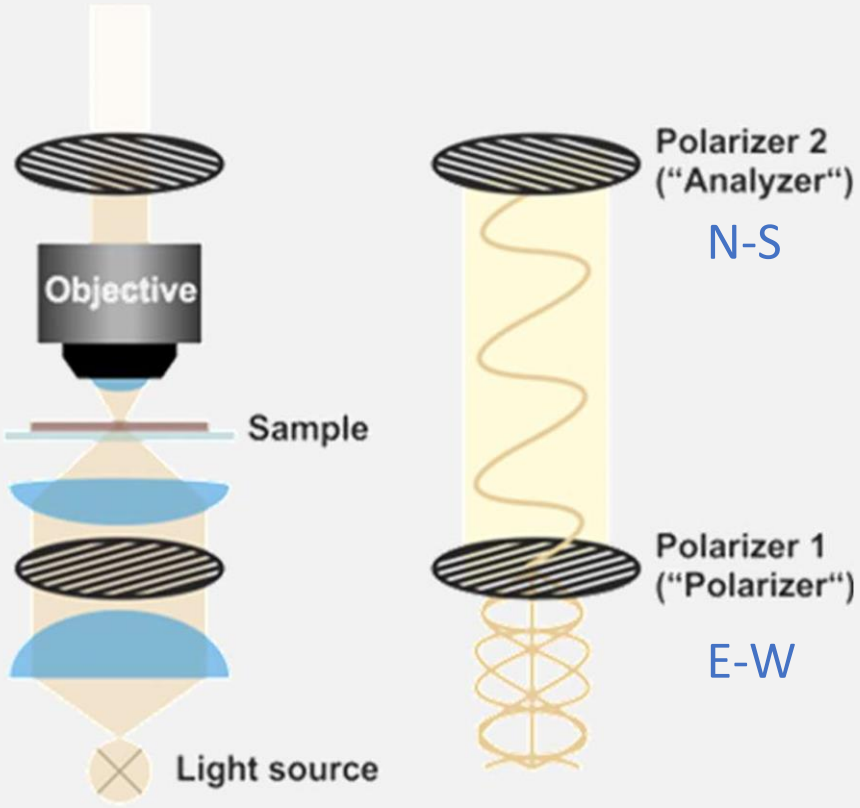
The petrological microscope: plane-polarized light (PPL)

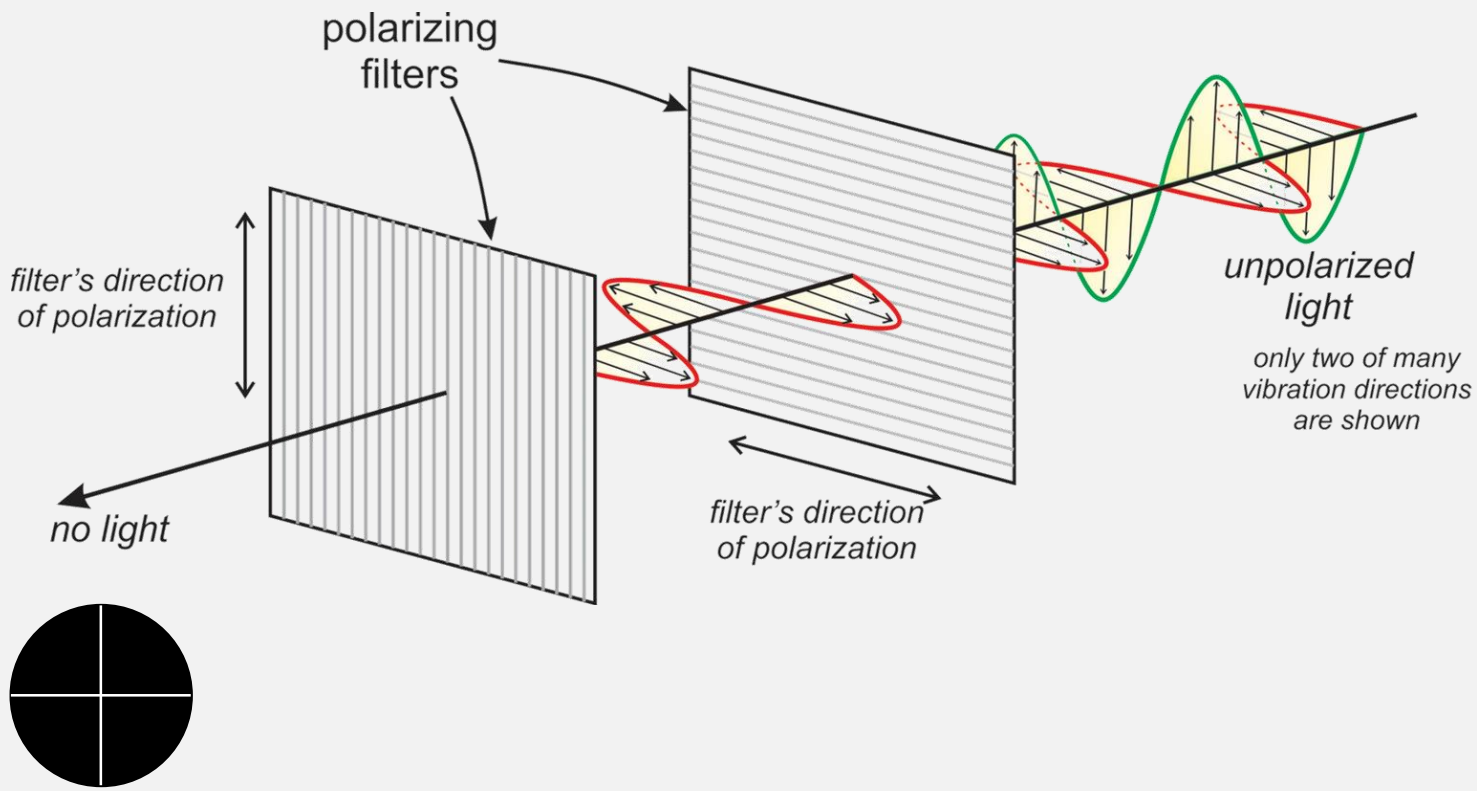




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The petrological microscope: crossed polars (XP)





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Optics and mineralogy

1.1 Light and the polarizing microscope

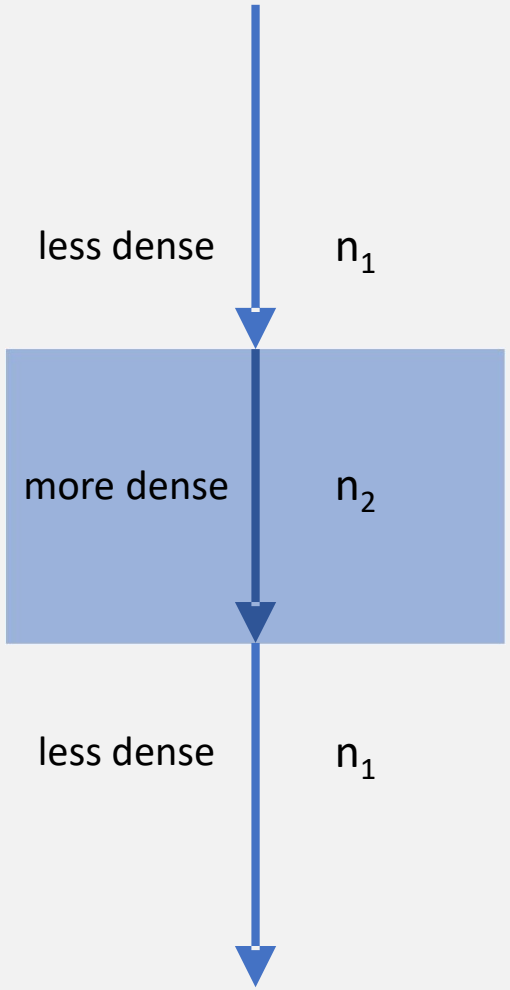
1.2 The refractive index

1.3 Relief in minerals



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The refractive index

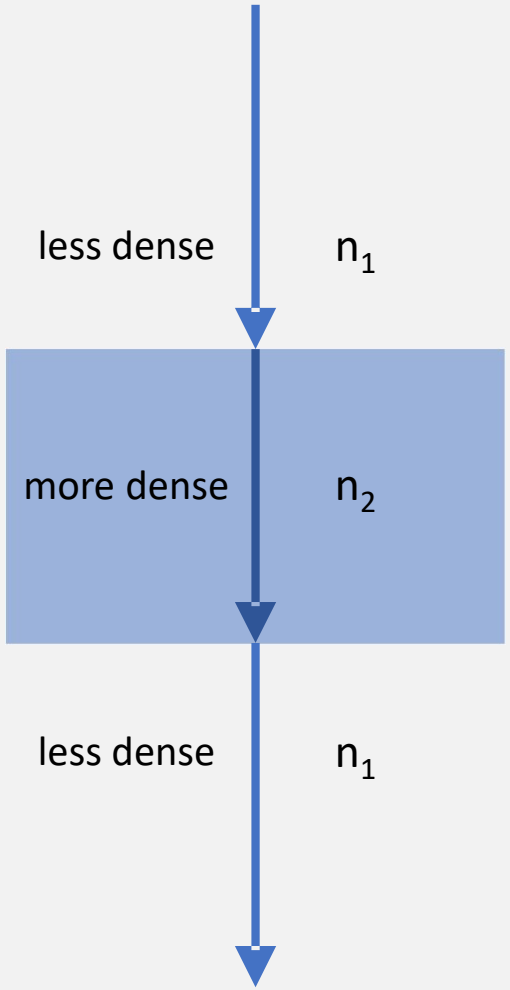


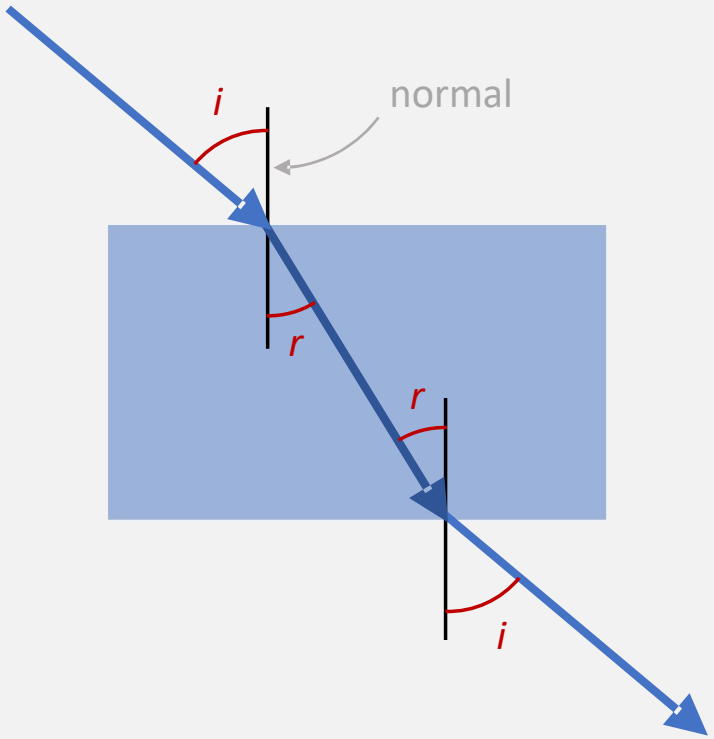
Refractive index:

$$RI = \frac{V_{air}}{V_{material}} = \frac{n_1}{n_2}$$

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The refractive index





Snell's Law:

$$\frac{\sin i}{\sin r} = \frac{V_{air}}{V_{material}}$$

$$\frac{\sin i}{\sin r} = RI = n$$

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Optics and mineralogy

1.1 Light and the polarizing microscope

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1.3 Relief in minerals

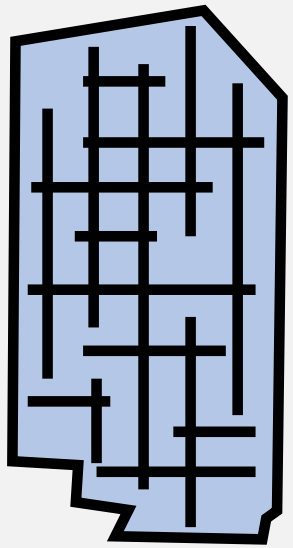


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Refractive index in minerals: Relief

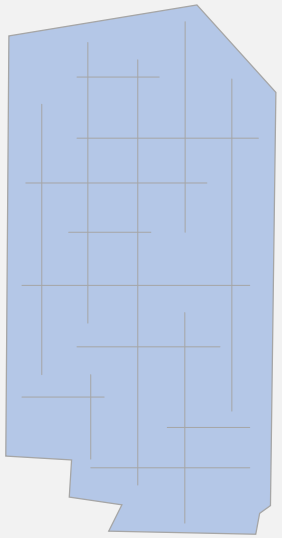
Relief is seen in plane-polarized light (PPL)

high relief



RI is very **different** to the thin section mounting medium

low relief



RI is very **similar** to the thin section mounting medium

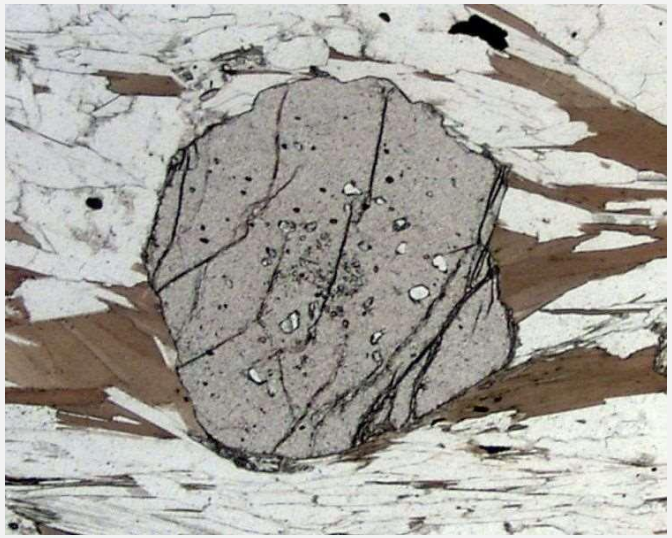
Thin section mounting medium has RI = 1.54

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Refractive index in minerals: Relief

Relief is seen in plane-polarized light (PPL)

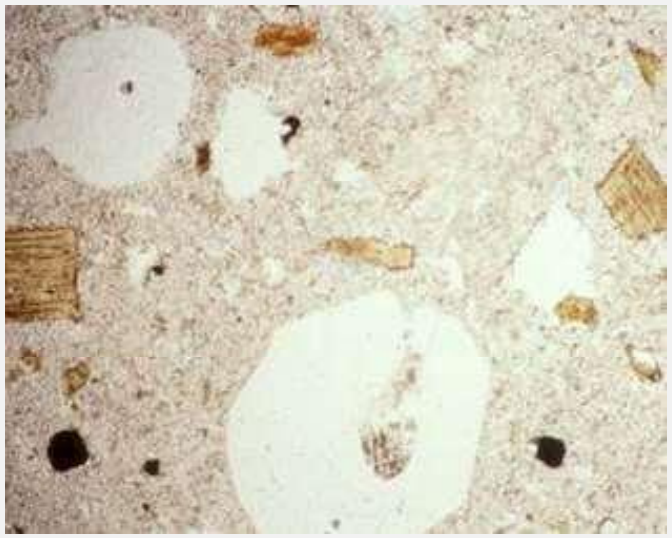
high relief



RI is very **different** to the thin section mounting medium

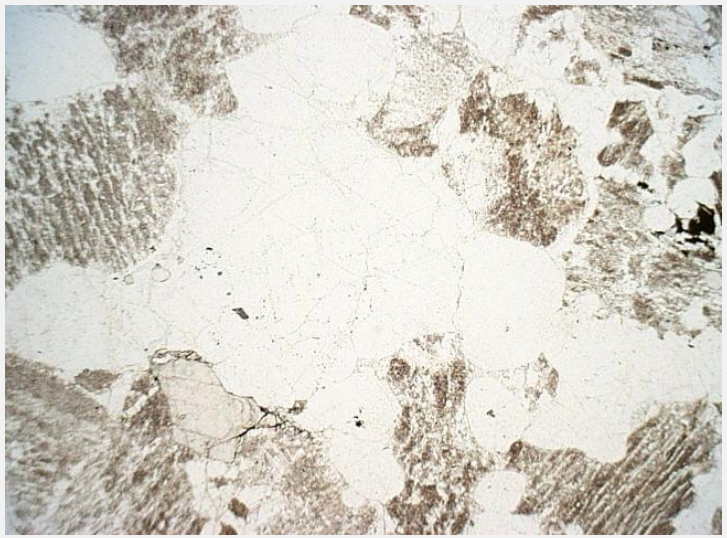
e.g. **garnet**: RI ≈ 0.8

low relief



RI is very **similar** to the thin section mounting medium

e.g. **quartz**: RI ≈ 1.55



Thin section mounting medium has RI = 1.54

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Optics and mineralogy

You should now be able to:

- Explain the concept of polarized light
- State the polarization directions of the polarizer and the analyzer on a petrological microscope
- Give a definition of the refractive index
- Use petrological observations to classify minerals as having low, medium or high relief



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Optics and mineralogy

2.1 The optical indicatrix

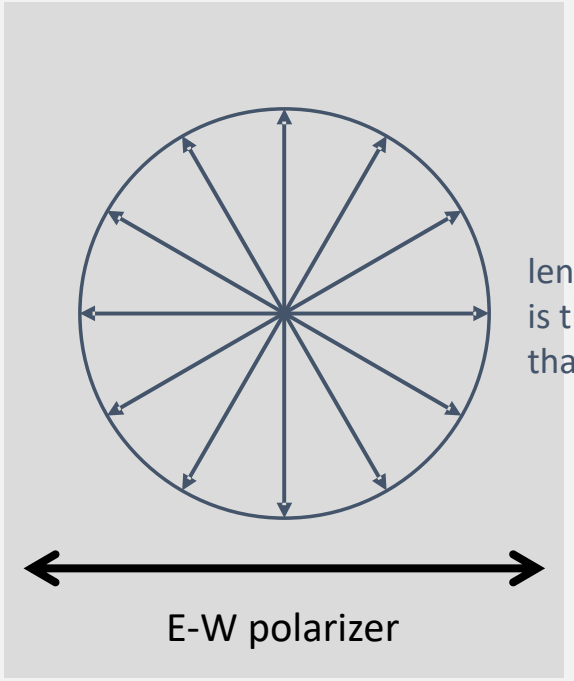
2.2 Crystal systems and the indicatrix



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The optical indicatrix: a construction for representing refractive indices in minerals

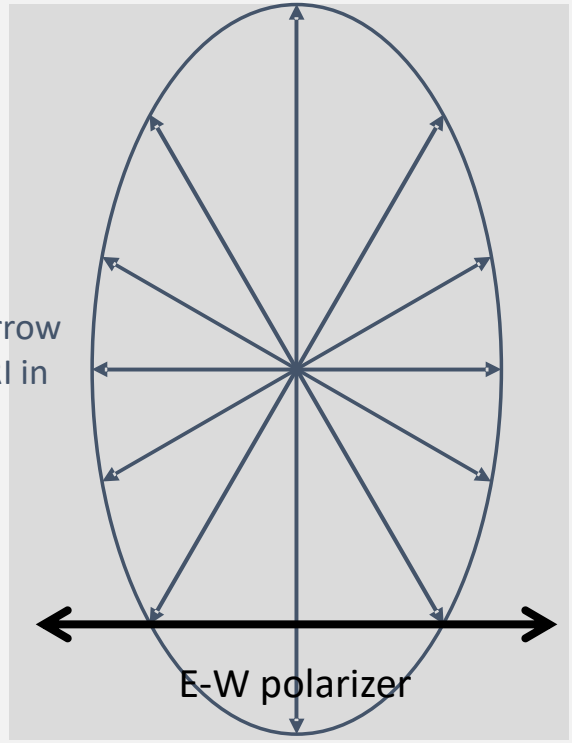
isotropic



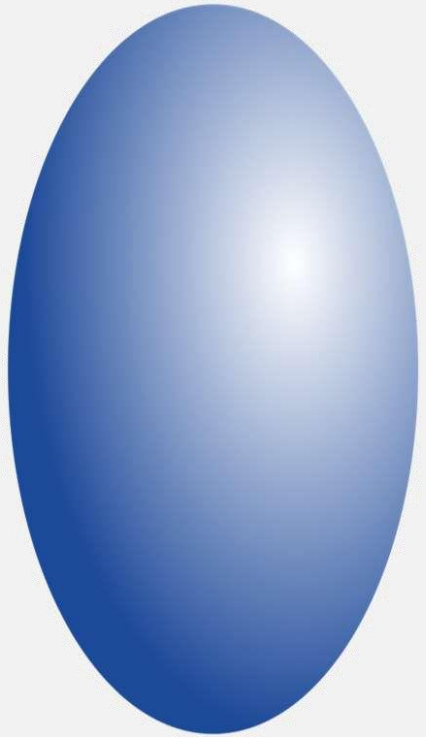
Same RI in all directions

anisotropic

length of blue arrow is the effective RI in that direction



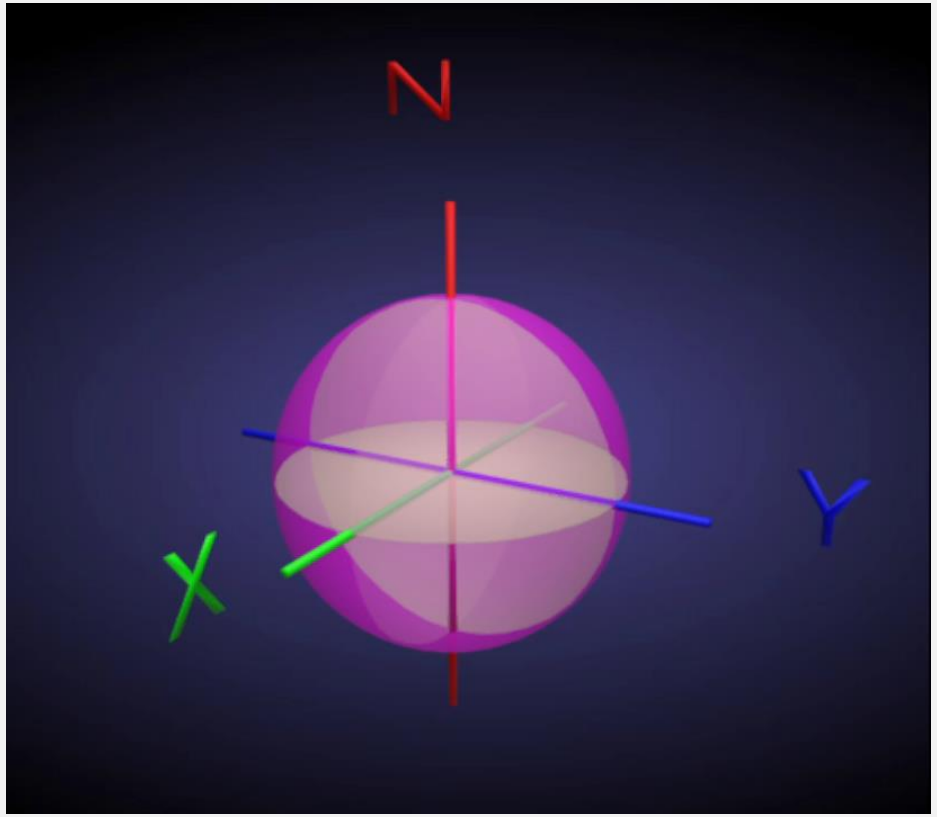
RI is different in each direction



In 3D: an ellipsoid called the **optical indicatrix**

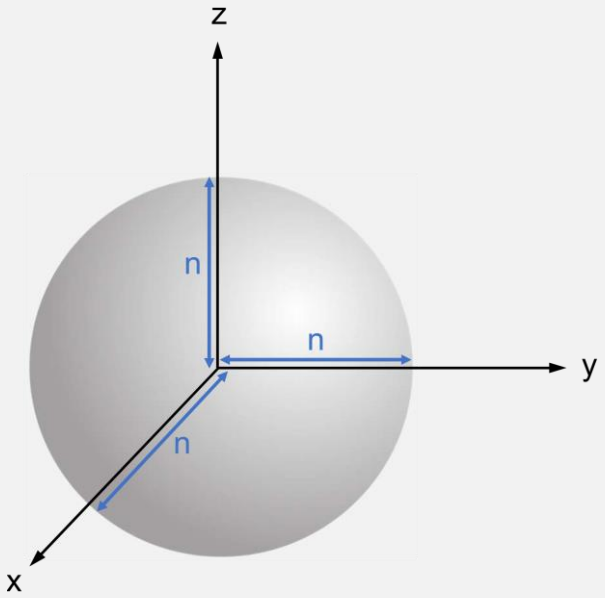
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Isotropic indicatrix



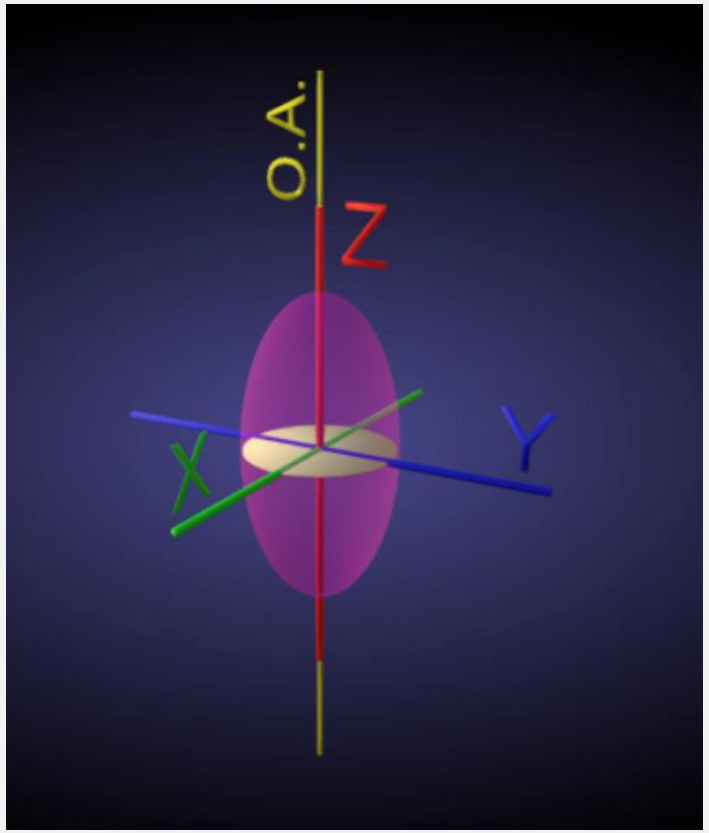
Minerals with one refractive index have an **isotropic** indicatrix

- The shape of the isotropic indicatrix is a **sphere**
- The refractive index is represented by the radius of the sphere, and is denoted ***n***
- 2D slices through the indicatrix are always circular, with radius *n*



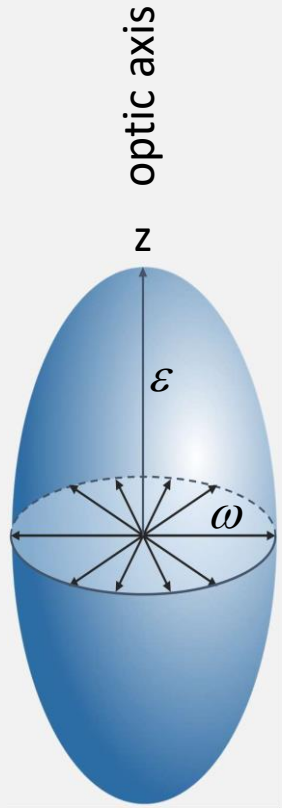
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Uniaxial indicatrix: one optic axis



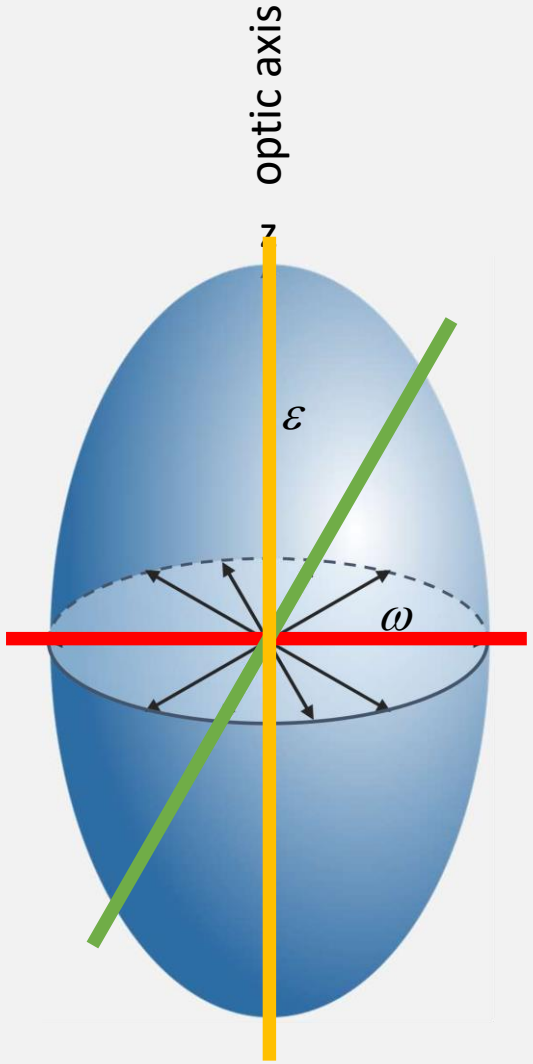
Minerals with two refractive indices have a **uniaxial** indicatrix

- The shape of the uniaxial indicatrix is an **ellipsoid of revolution**
- The refractive indices are represented by the semi-major and semi-minor axes of the ellipsoid, and are denoted **ω** and **ε**
- ω and ε are always at 90° to each other
- There is one 2D slice that gives a circular cross-section. The circular section always sits in the crystallographic x-y plane. It has radius ω
- The optic axis is defined as the direction we look down to see the circular section (i.e., the optic axis is perpendicular to the circular section)
- The optic axis is always parallel to the crystallographic z axis
- ε is always parallel to the z axis and the optic axis
- In these diagrams, ε > ω



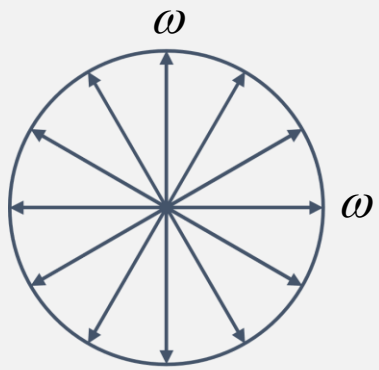
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Uniaxial indicatrix: one optic axis

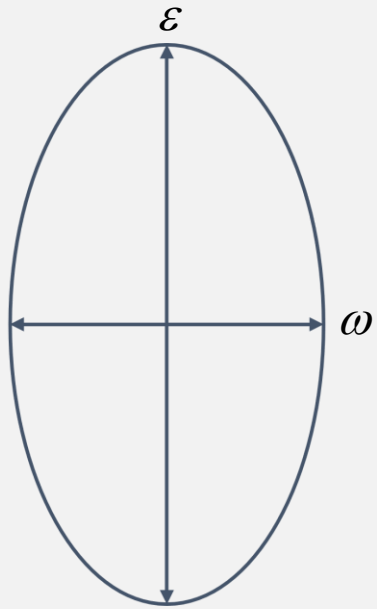


Possible slices through the uniaxial indicatrix

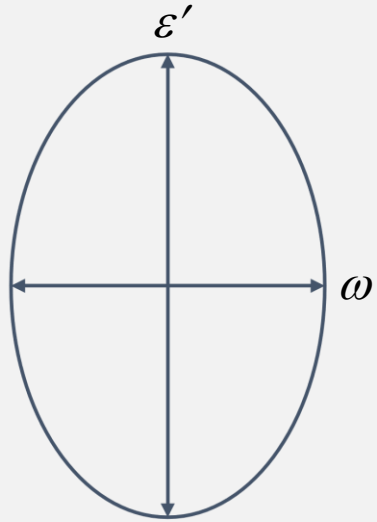
2D slice perpendicular to the optic axis: circular section



2D slice parallel to the optic axis



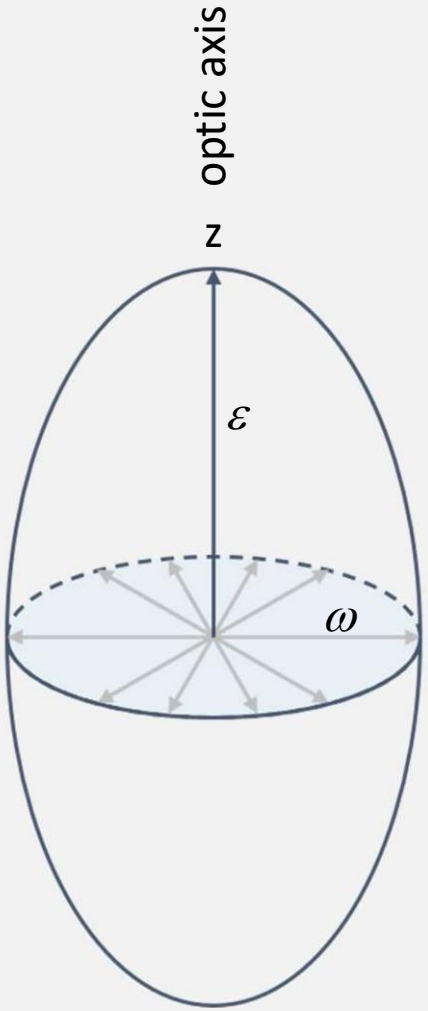
2D slice at a random angle to the optic axis



ε′ is intermediate between ω and ε

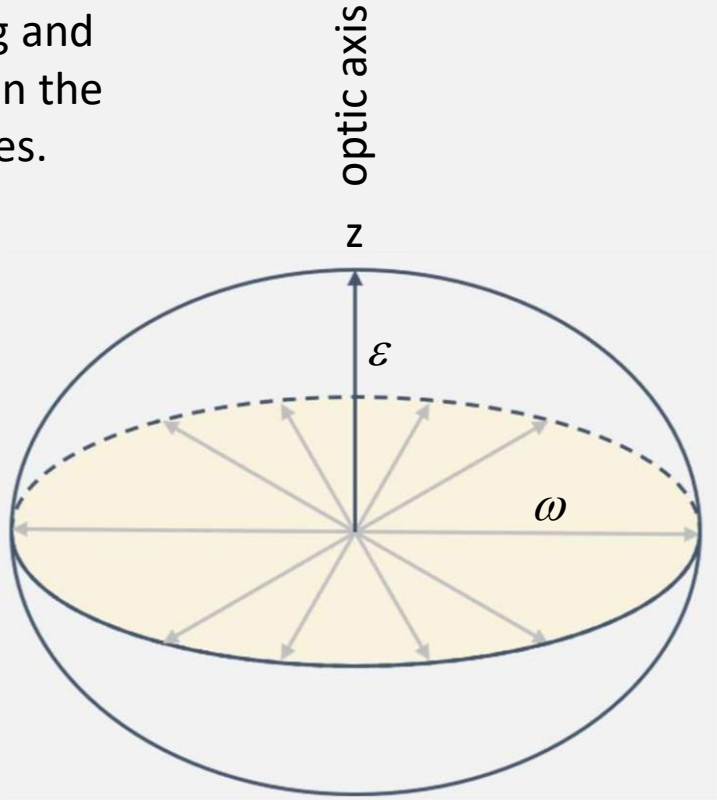
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Uniaxial indicatrix: one optic axis



The uniaxial indicatrix can be prolate (long and thin) or oblate (short and fat) depending on the relative sizes of the two refractive indices.

Prolate: $\varepsilon > \omega$

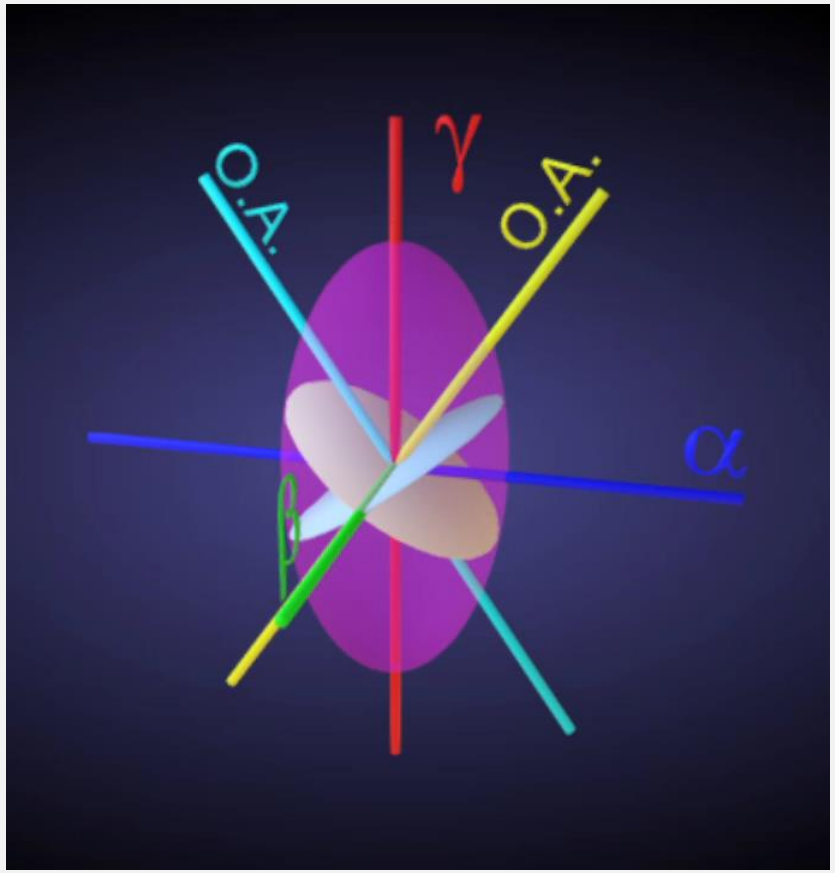


Oblate: $\varepsilon < \omega$

Light travels fastest in the direction of whichever RI is the shortest.

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Biaxial indicatrix: two optic axes

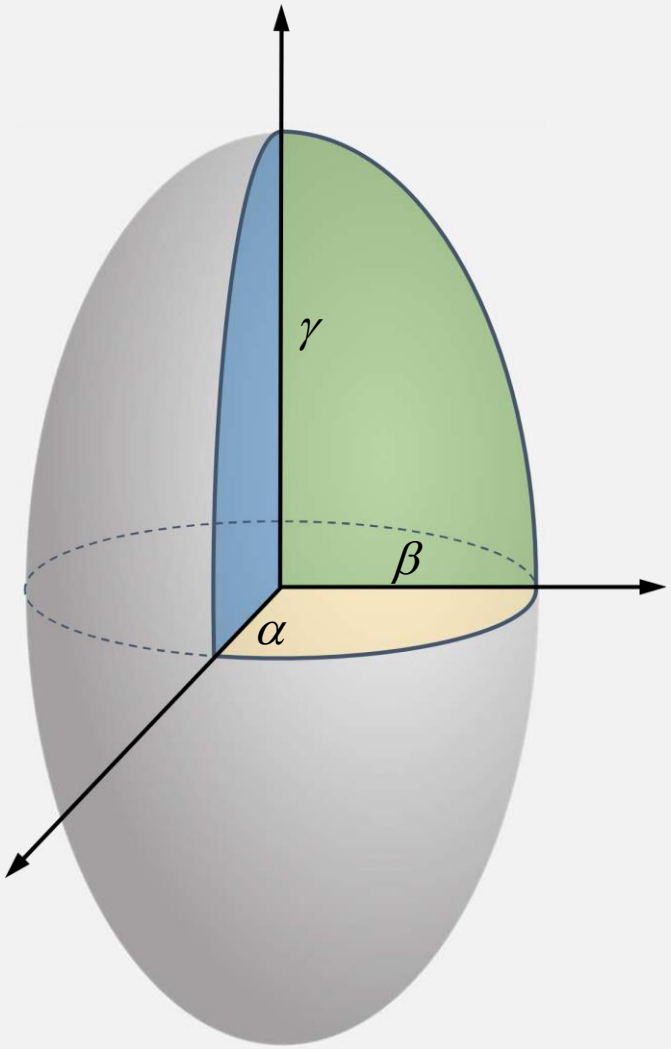


Minerals with three refractive indices have a biaxial indicatrix

- The shape of the biaxial indicatrix is a **general ellipsoid**
- The refractive indices are represented by the lengths of the semi-axes of the ellipsoid, and are denoted α, β and γ
- α, β and γ are always at 90° to each other
- By convention, $\alpha < \beta < \gamma$. Light travels fastest along α and slowest along γ
- There are two possible 2D slices that give circular cross-sections. The circular sections both have radius β
- The optic axis is defined as the direction we look down to see the circular sections (i.e., the optic axes are perpendicular to the circular sections)

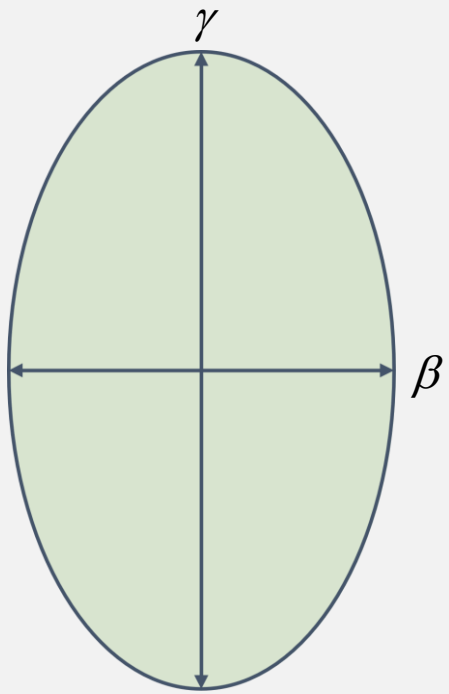
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Biaxial indicatrix: two optic axes

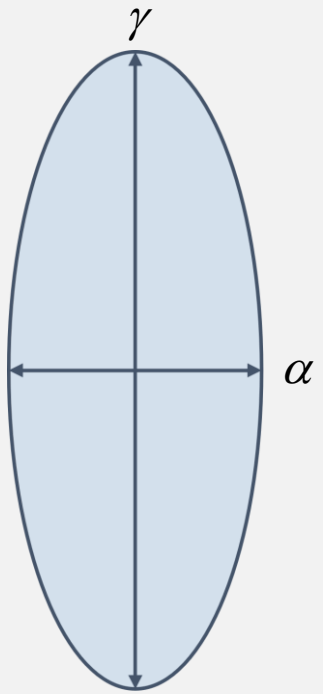


Possible slices through the biaxial indicatrix

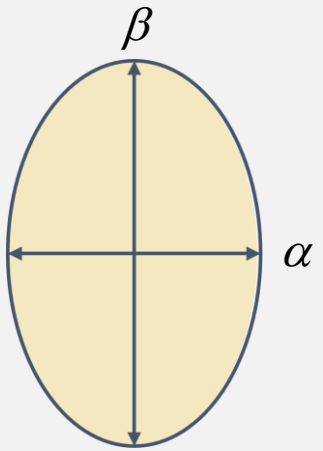
view down α



view down β

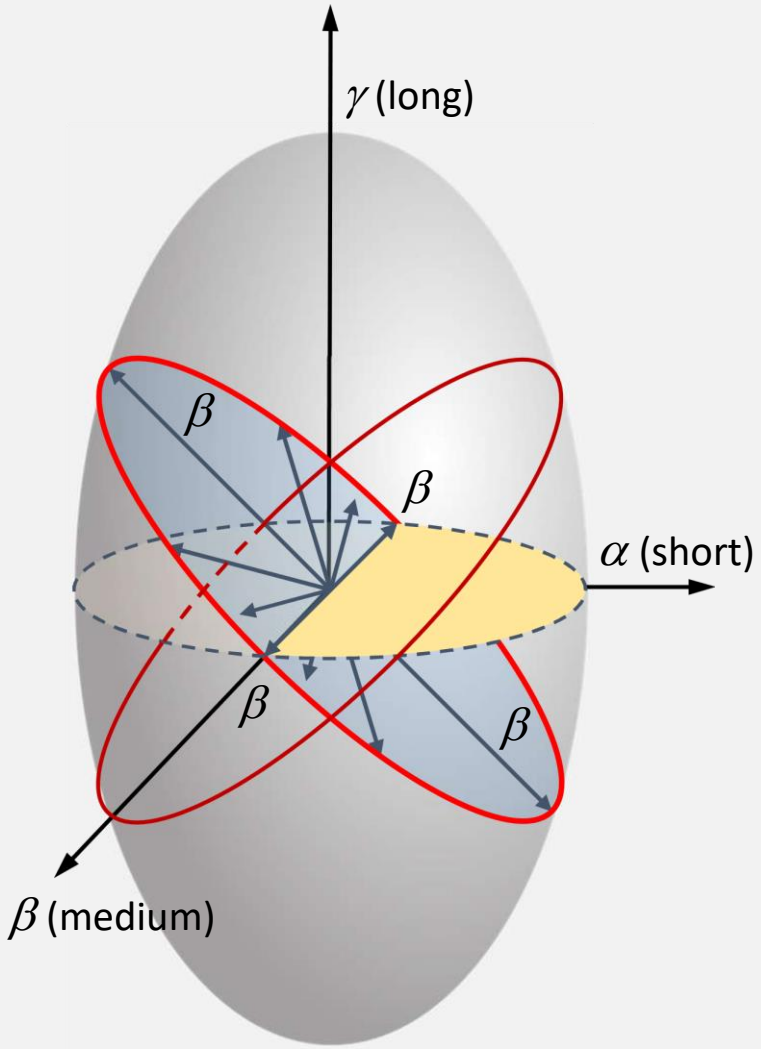


view down γ



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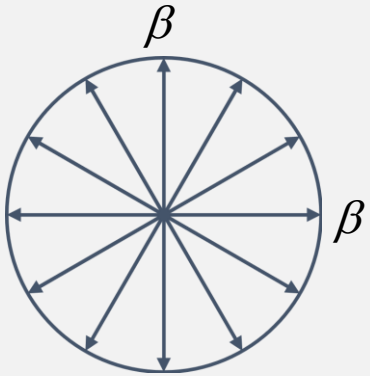
Biaxial indicatrix: two optic axes



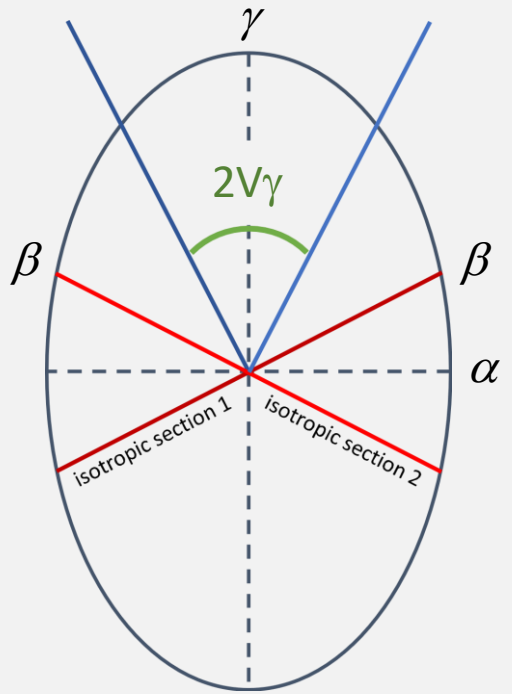
Possible slices through the biaxial indicatrix

Two circular sections with radius β

view down either optic axis:
circular section



optic axis 1 optic axis 2



View down β is the α-γ plane, which is end-on to both the circular sections

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Optics and mineralogy

2.1 The optical indicatrix

2.2 Crystal systems and the indicatrix



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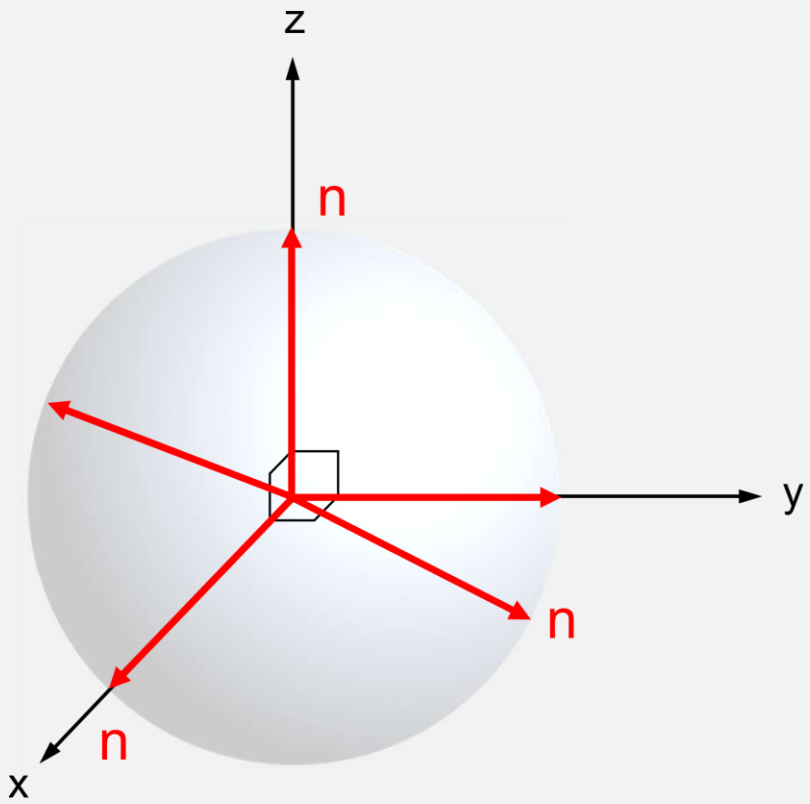
Crystal system	Indicatrix classification	Defining RIs	Orientation of RIs with respect to *x*, *y* and *z* axes
Cubic	isotropic	n	No orientation constraints
Trigonal (rhombohedral)	uniaxial	ε and ω	ε is parallel to *z*
Tetragonal			
Hexagonal			
Orthorhombic	biaxial	α, β and γ	α, β and γ are parallel to *x*, *y* and *z* (in any order)
Monoclinic			One of α, β or γ is parallel to *y*
Triclinic			No orientation constraints

The shape and orientation of the indicatrix must obey the crystallographic symmetry of the mineral



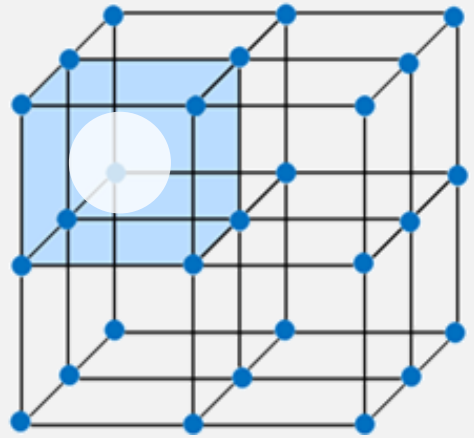
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Optical symmetry matches the crystallographic symmetry: Isotropic minerals



Isotropic indicatrix: Fits with **cubic** symmetry

The isotropic indicatrix can sit in any orientation inside the cubic unit cell



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Optical symmetry matches the crystallographic symmetry: Uniaxial minerals

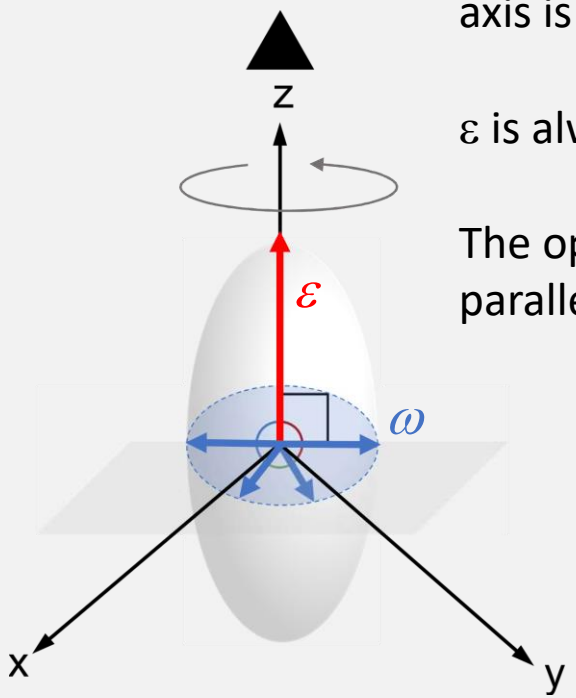
Uniaxial indicatrix: Fits the three crystal systems with **high degrees of rotational symmetry**

trigonal

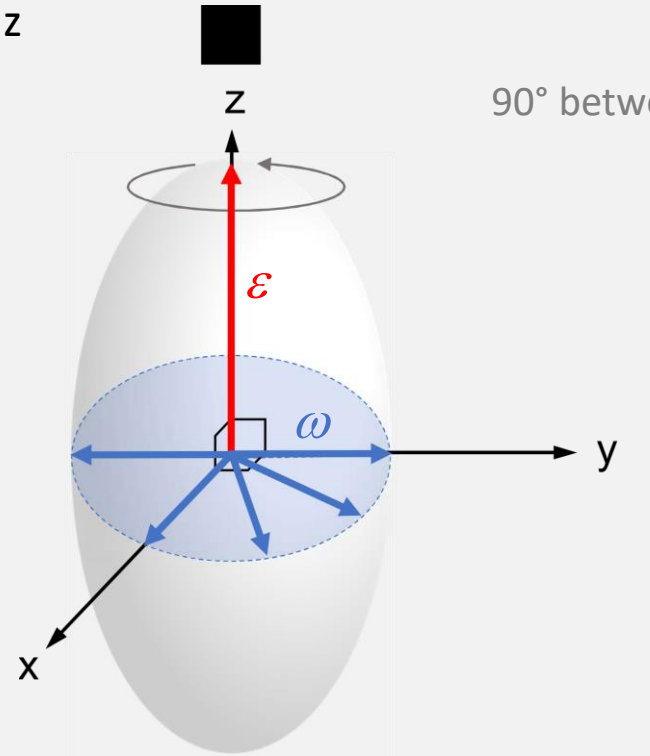
The high-order rotation axis is always parallel to z

ε is always parallel to z

The optic axis is always parallel to z

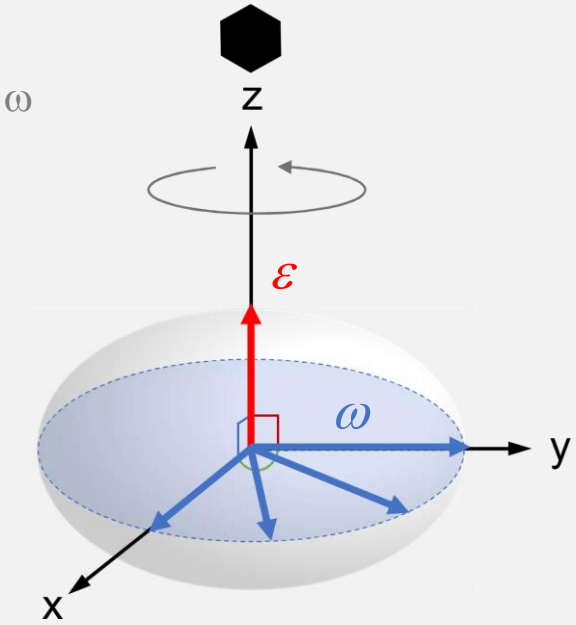


tetragonal



90° between ε and ω

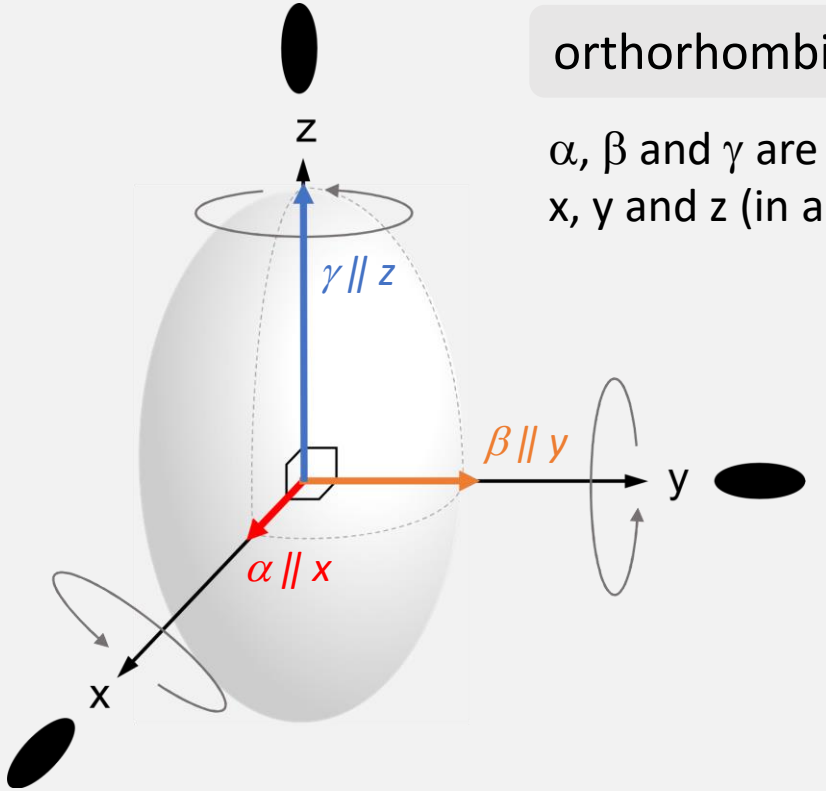
hexagonal



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Optical symmetry matches the crystallographic symmetry: Biaxial minerals

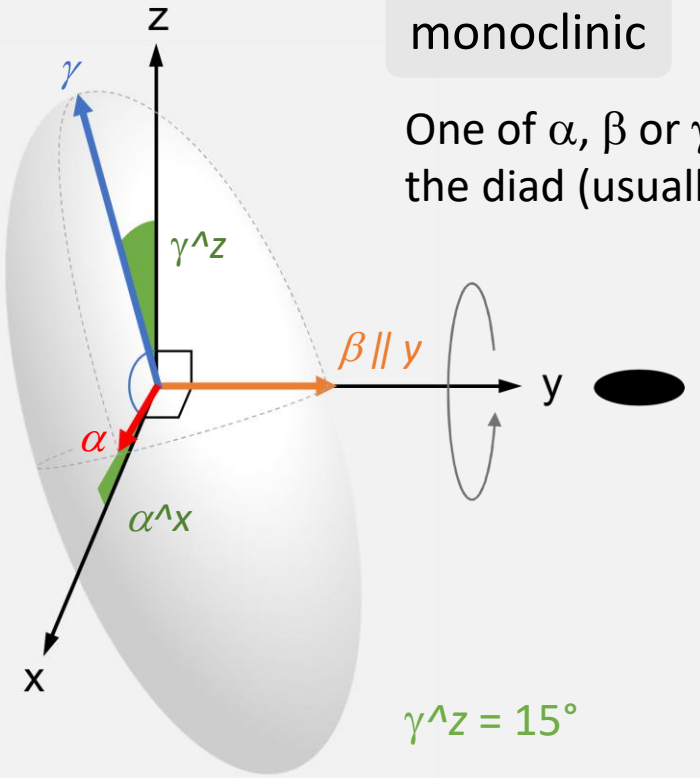
Biaxial indicatrix: Fits the two crystal systems with **lower degrees of symmetry**



orthorhombic

α, β and γ are parallel to x, y and z (in any order)

90° between α, β and γ



monoclinic

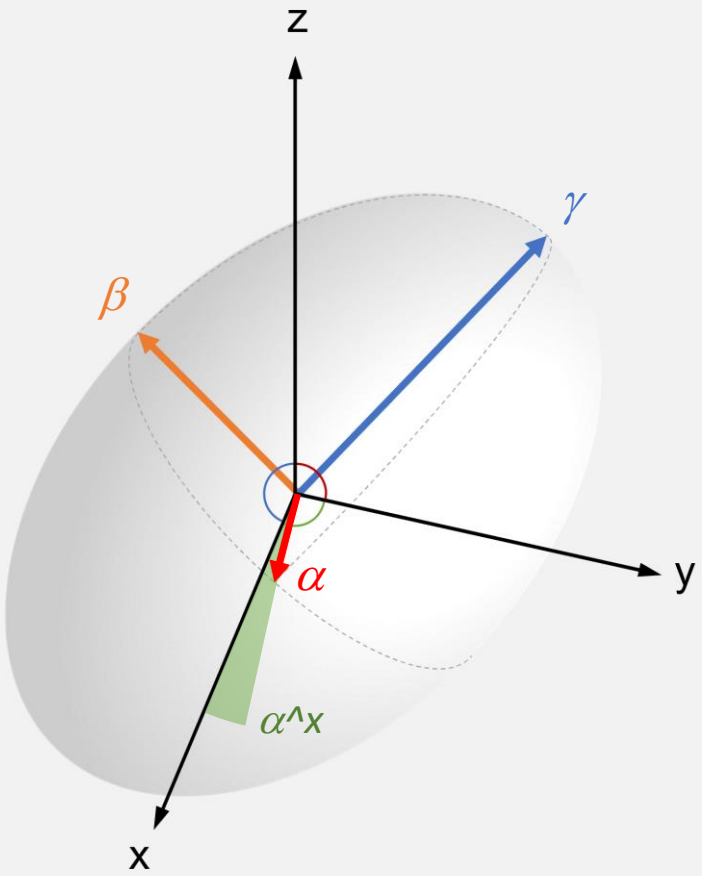
One of α, β or γ is parallel to the diad (usually the y axis)

γ^z = 15°

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Optical symmetry matches the crystallographic symmetry: Biaxial minerals

Biaxial indicatrix: Fits the crystal system with **no symmetry constraints**



triclinic

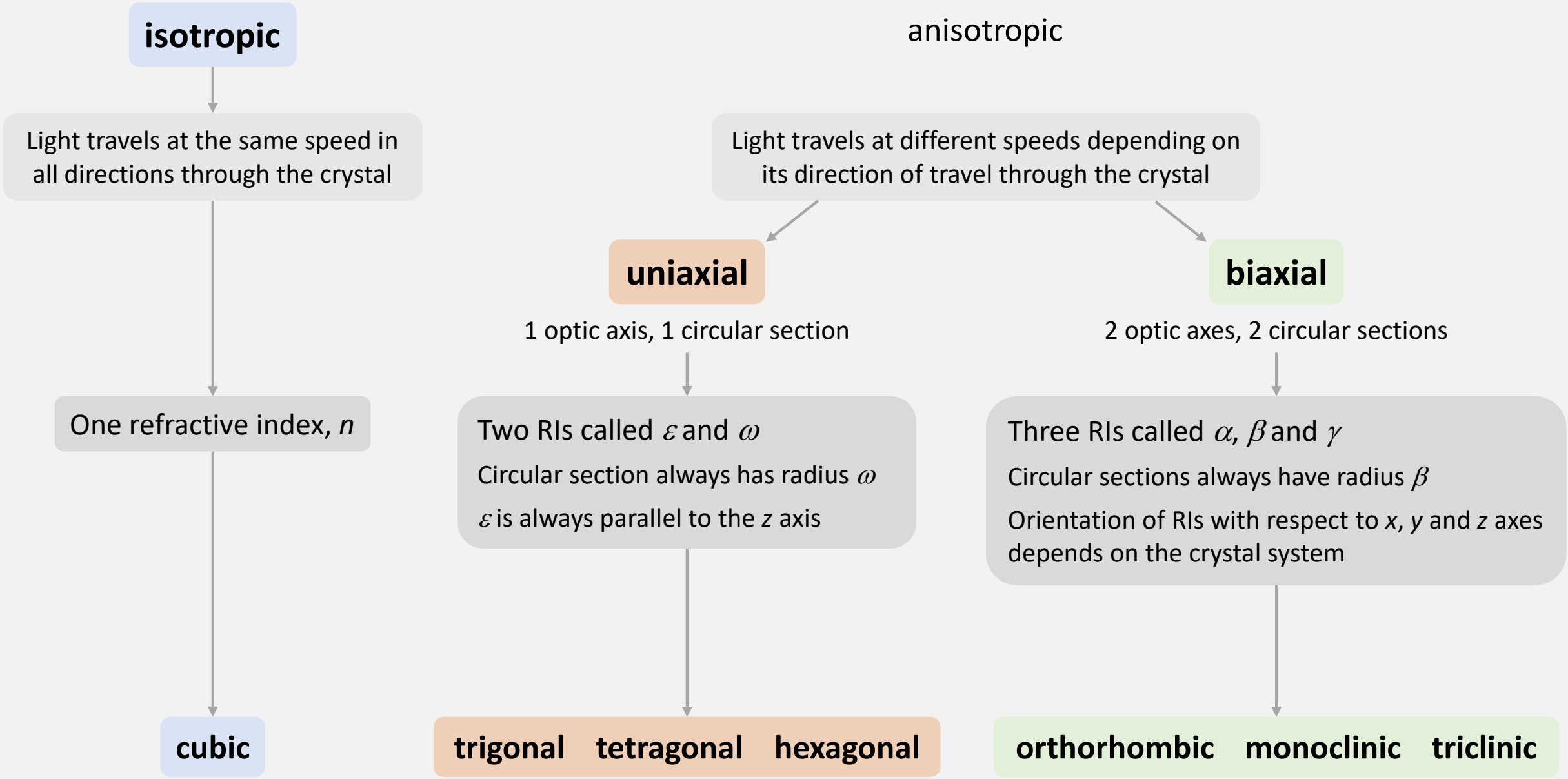
No orientation constraints: α, β and γ do not have to be aligned with any of the crystallographic axes

The angle between a refractive index and a crystallographic axis is denoted with the ^ symbol

- Example: α^x denotes the angle between α and x

90° between α, β and γ

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Optics and mineralogy

You should now be able to:

- Explain the concept of the optical indicatrix
- Explain the relationship between the number of refractive indices, the shape of the indicatrix, and the underlying crystal symmetry
- Assign the correct indicatrix shape and number of refractive indices to each crystal system



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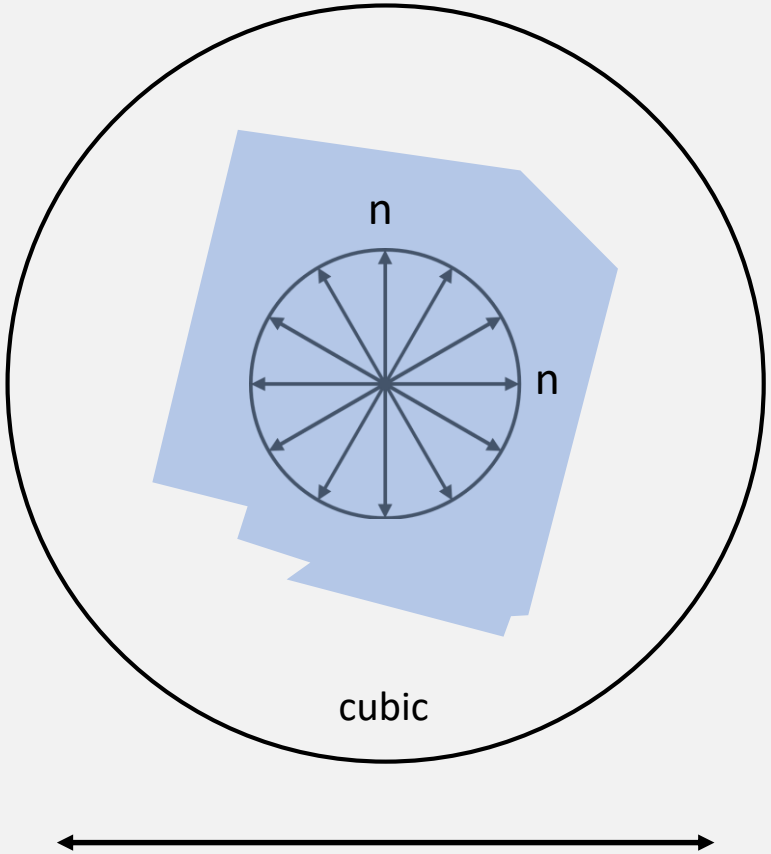
Optics and mineralogy

3.1 Optical properties of isotropic minerals

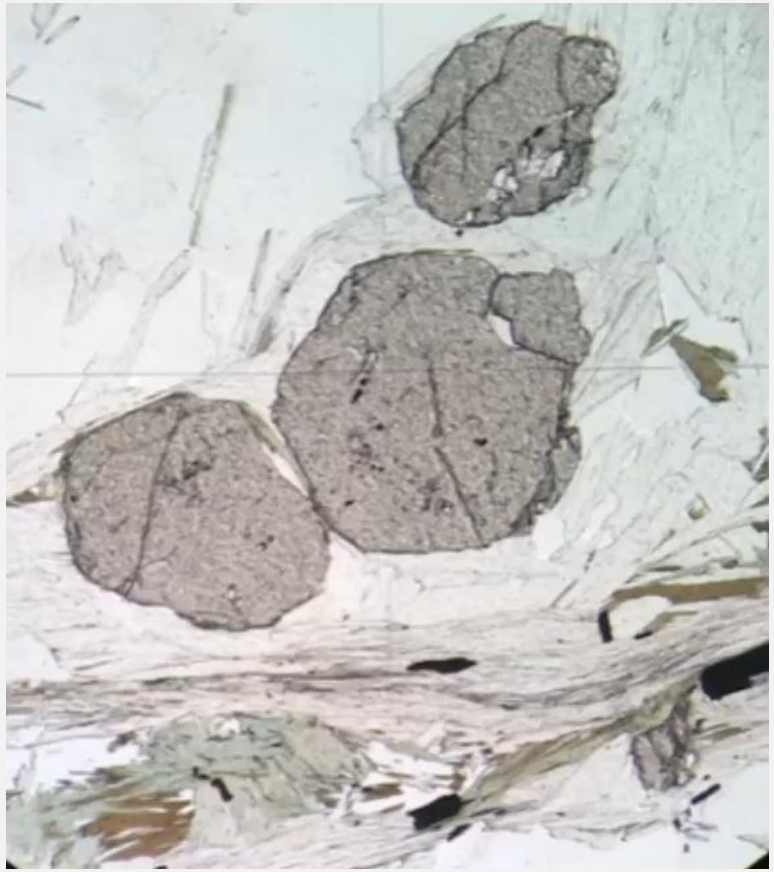


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Isotropic minerals in PPL



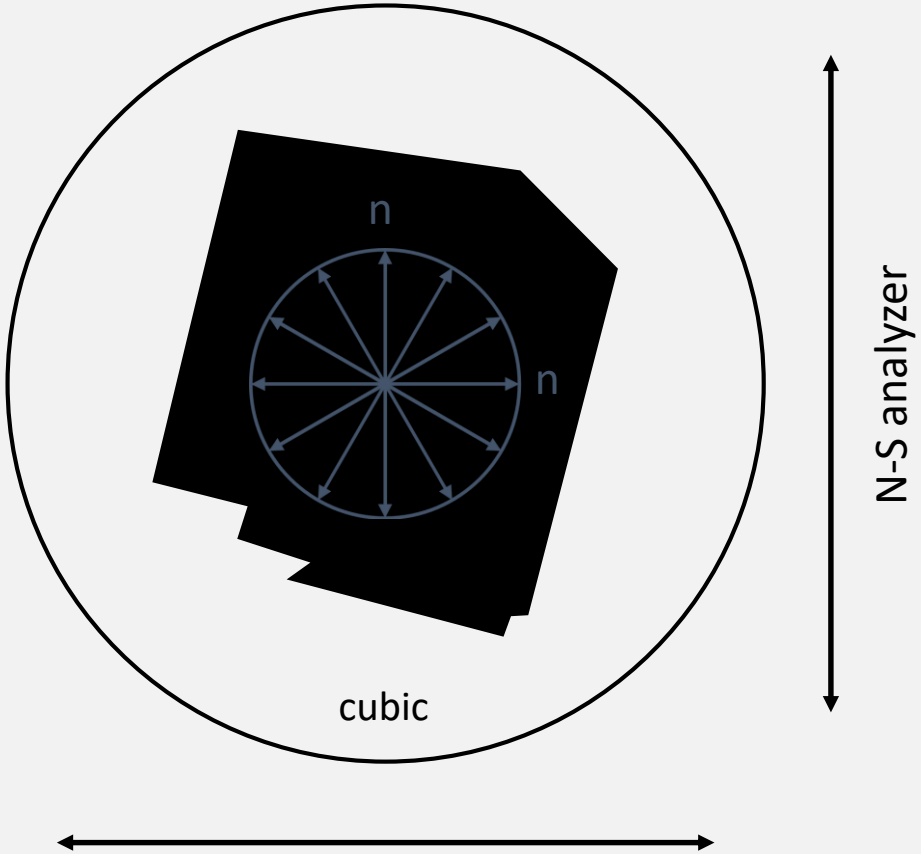
plane-polarized light



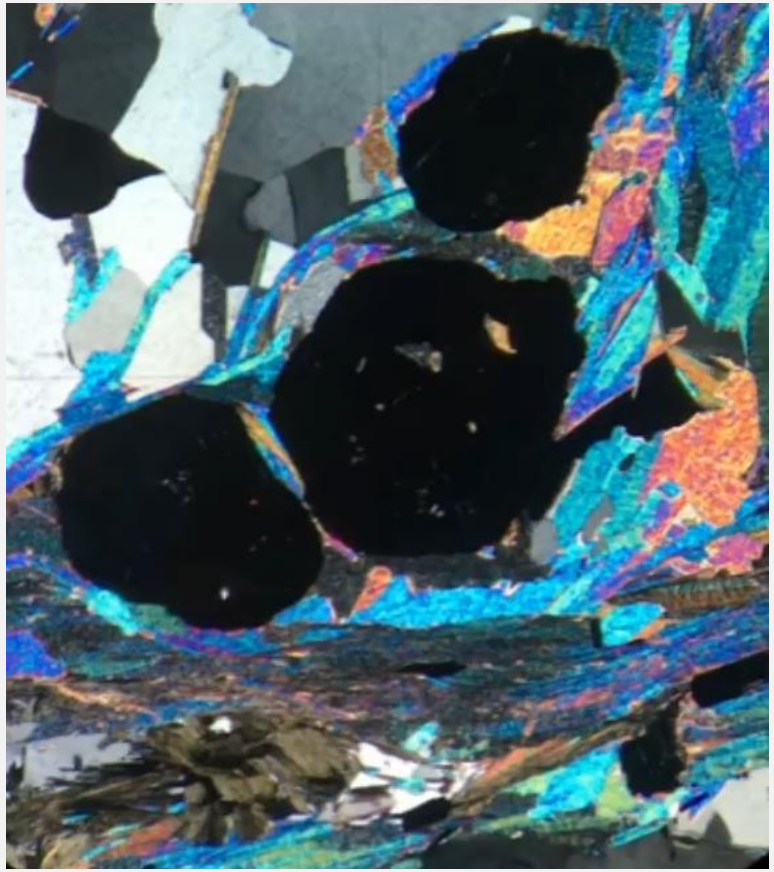
garnet: cubic

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Isotropic minerals in XP



crossed polars



garnet: cubic

extinction

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Optics and mineralogy

You should now be able to:

- Recognise isotropic minerals in thin section from their optical properties
- Explain why isotropic minerals are always extinct under crossed polars



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Optics and mineralogy

Optical properties of anisotropic minerals in plane-polarized light

4.1 Pleochroism in uniaxial minerals

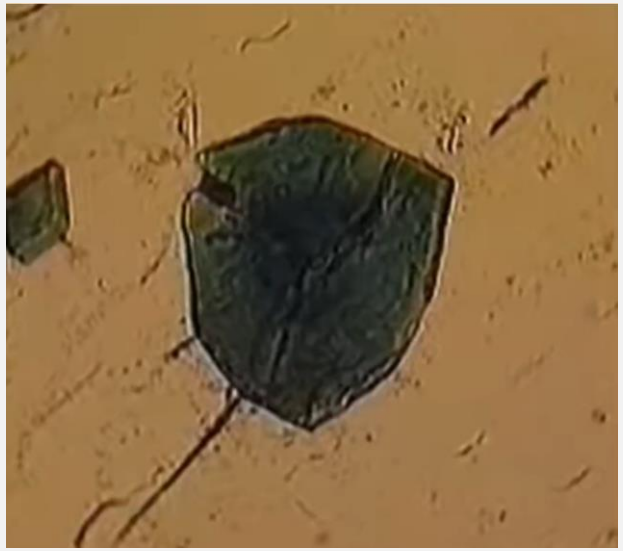
4.2 Pleochroism in biaxial minerals



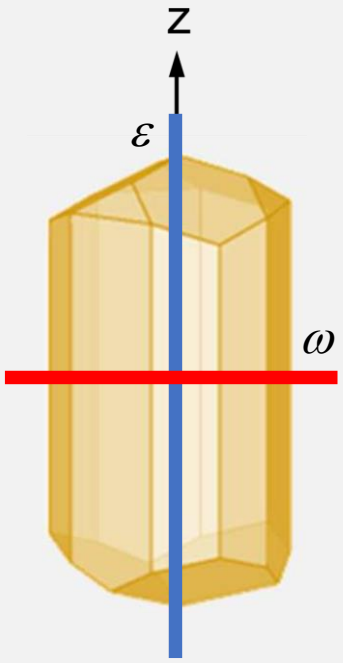
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Anisotropic minerals in PPL: Uniaxial

optic axis (basal) section

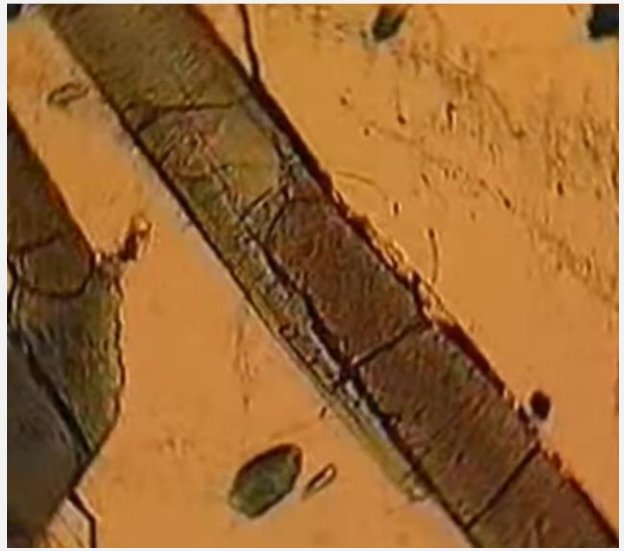


E-W polarizer



tourmaline
trigonal (–)

prismatic section



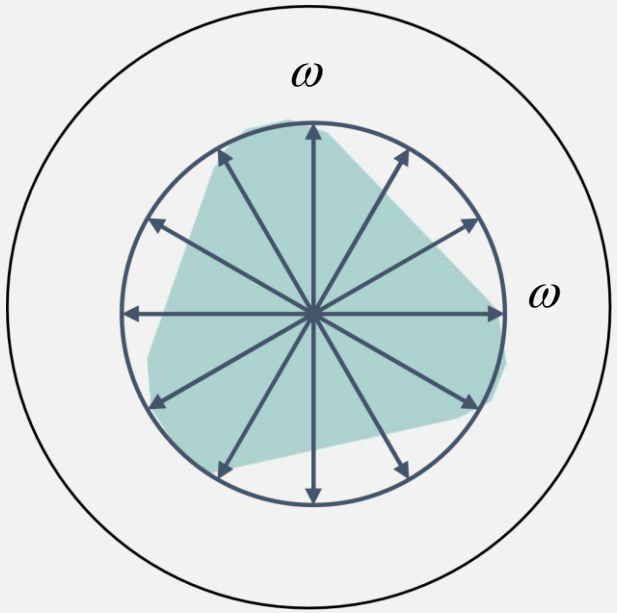
E-W polarizer

pleochroism

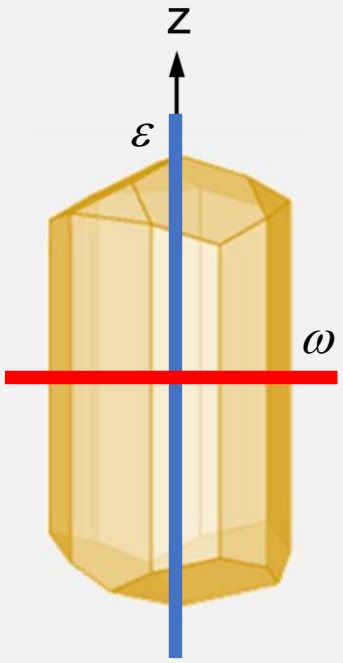
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Anisotropic minerals in PPL: Uniaxial

optic axis (basal) section

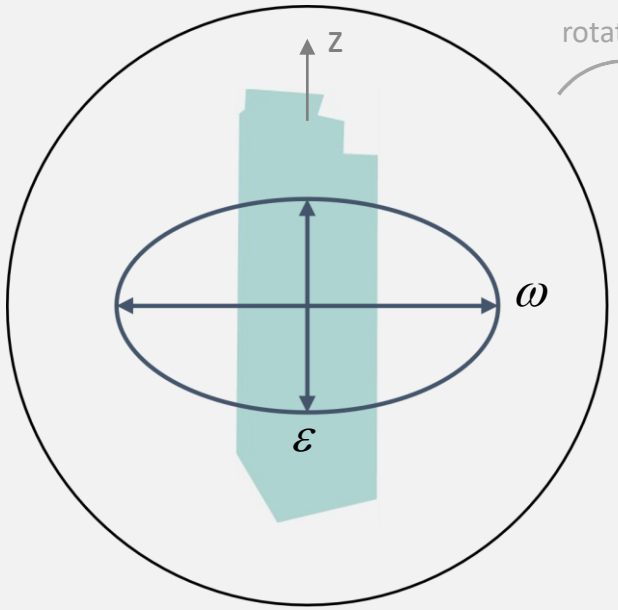


E-W polarizer

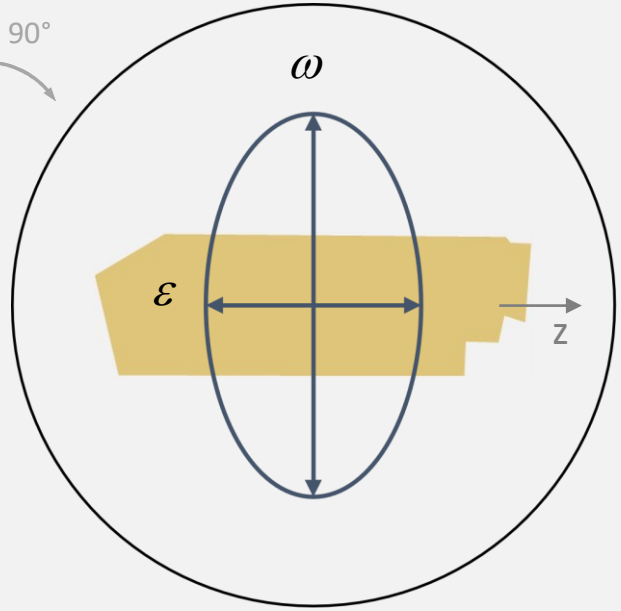


tourmaline
trigonal (–)

prismatic section



E-W polarizer

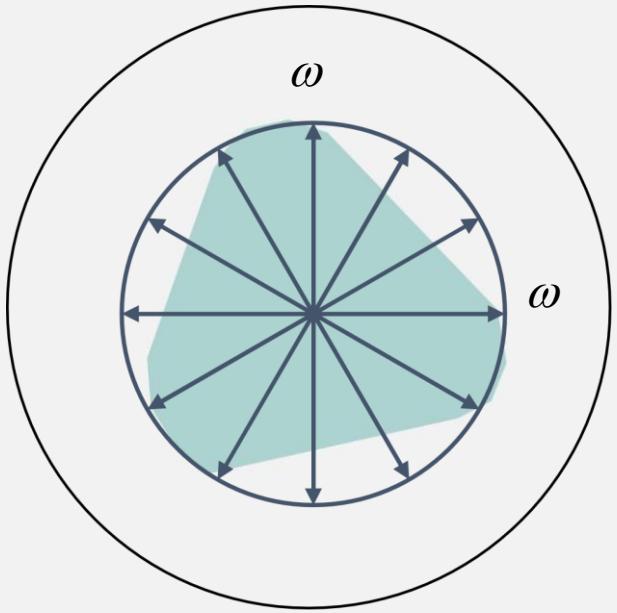


pleochroic scheme:
ω = blue-green
ε = yellow

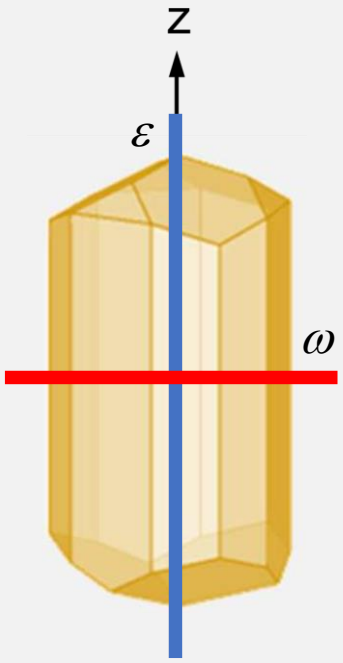
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Anisotropic minerals in PPL: Uniaxial

optic axis (basal) section

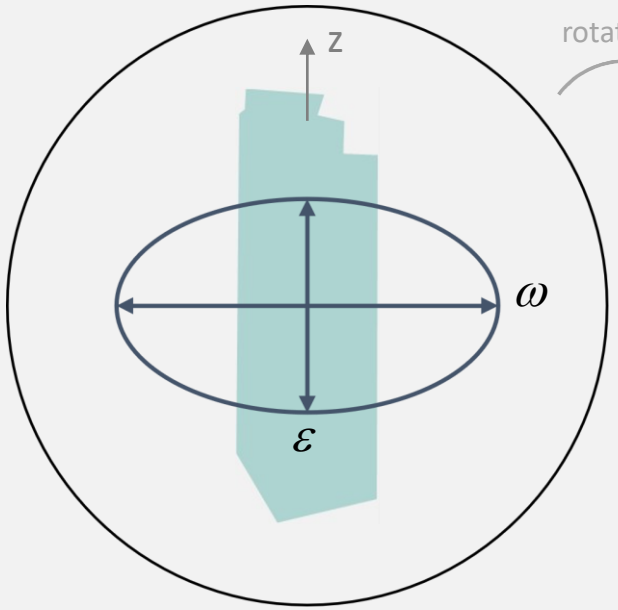


E-W polarizer

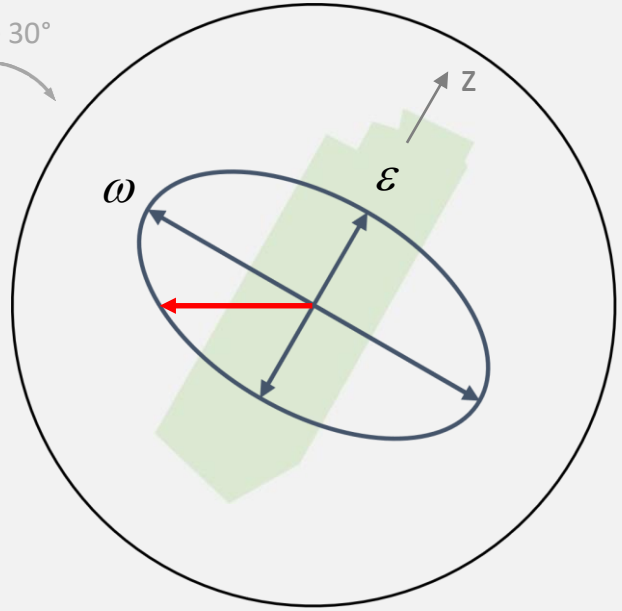


tourmaline
trigonal (–)

prismatic section



E-W polarizer

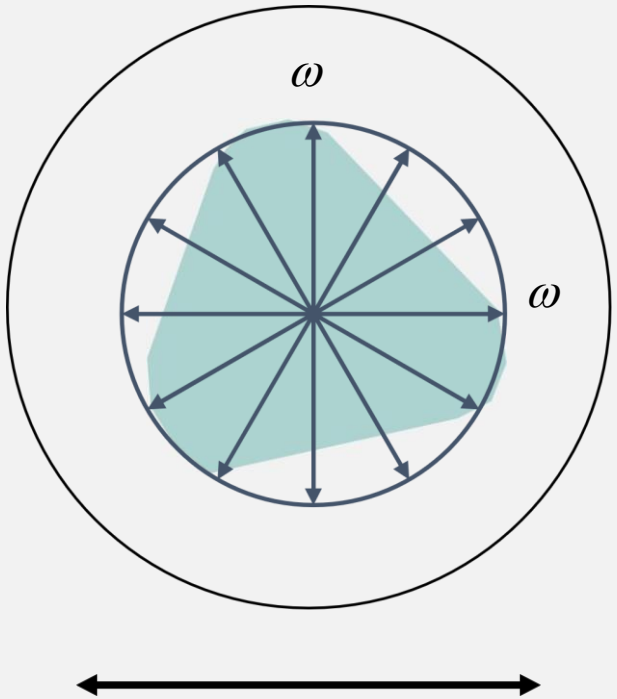


pleochroic scheme:
ω = blue-green
ε = yellow

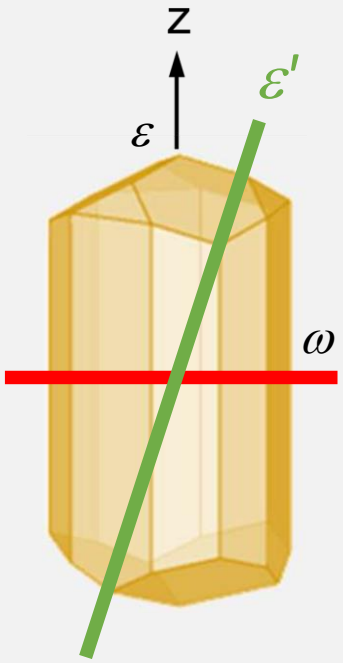
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Anisotropic minerals in PPL: Uniaxial

optic axis (basal) section

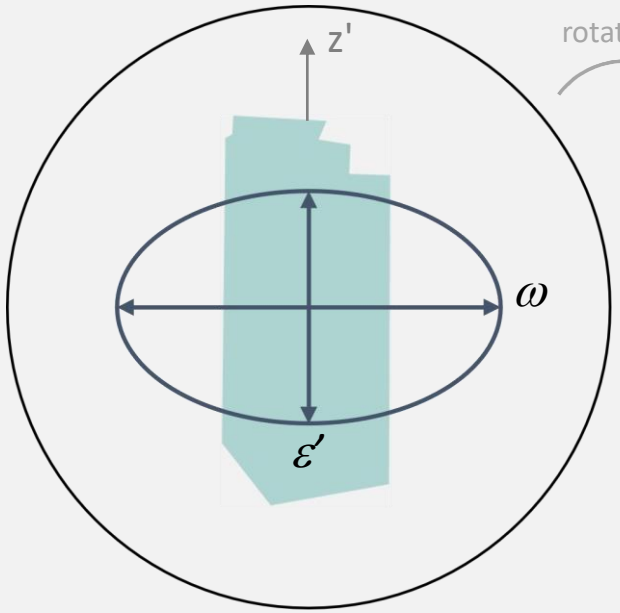


E-W polarizer

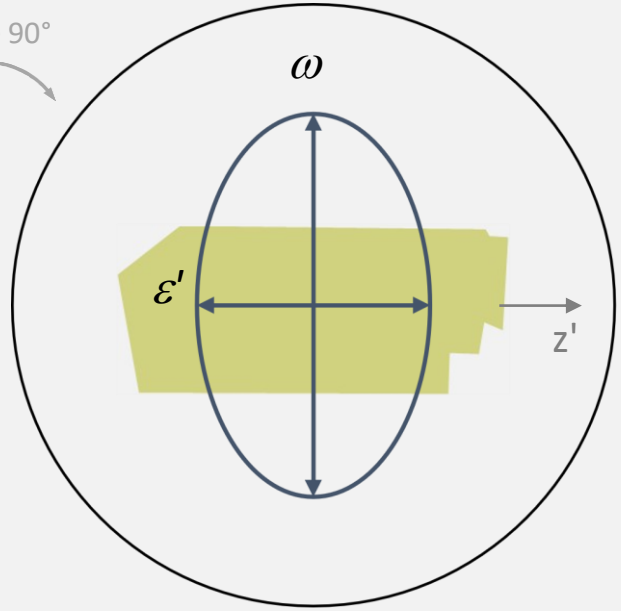


tourmaline
trigonal (–)

angled section



E-W polarizer



pleochroic scheme:
ω = blue-green
ε = yellow

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Optics and mineralogy

Optical properties of anisotropic minerals in plane-polarized light

4.1 Pleochroism in uniaxial minerals

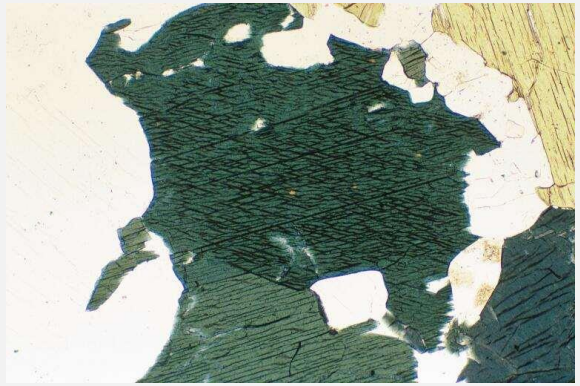
4.2 Pleochroism in biaxial minerals

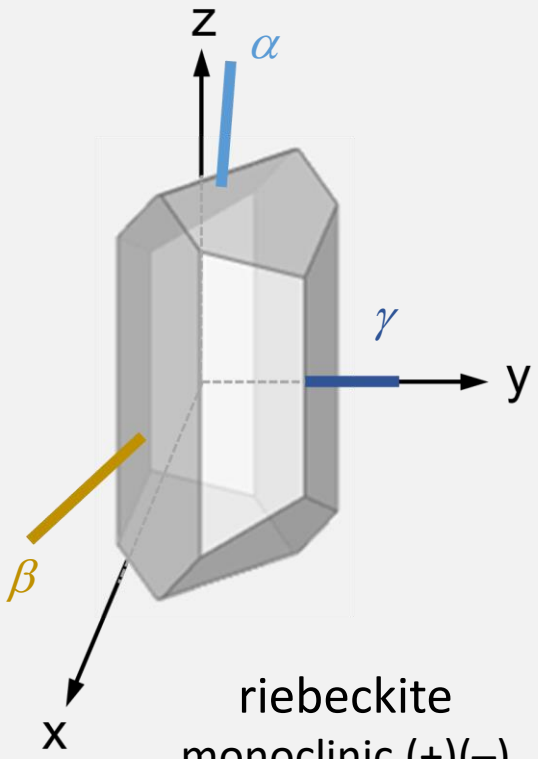


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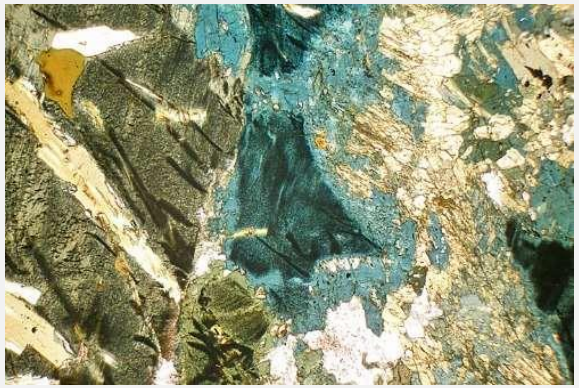
Anisotropic minerals in PPL: Biaxial

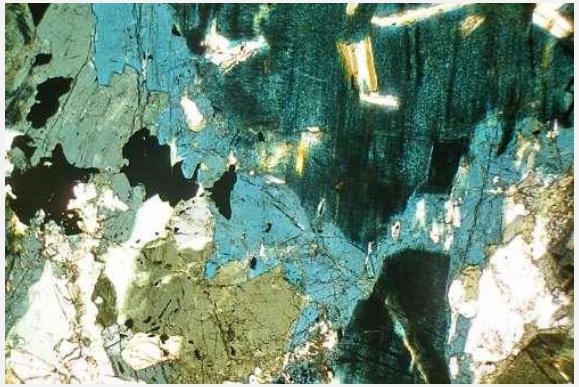






riebeckite
monoclinic (+)(–)



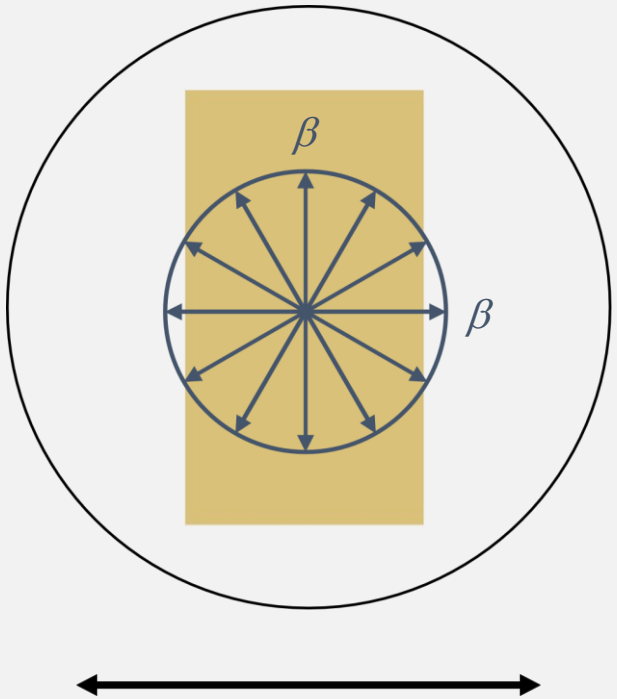


pleochroic scheme:
α = blue
β = yellow-brown
γ = dark blue

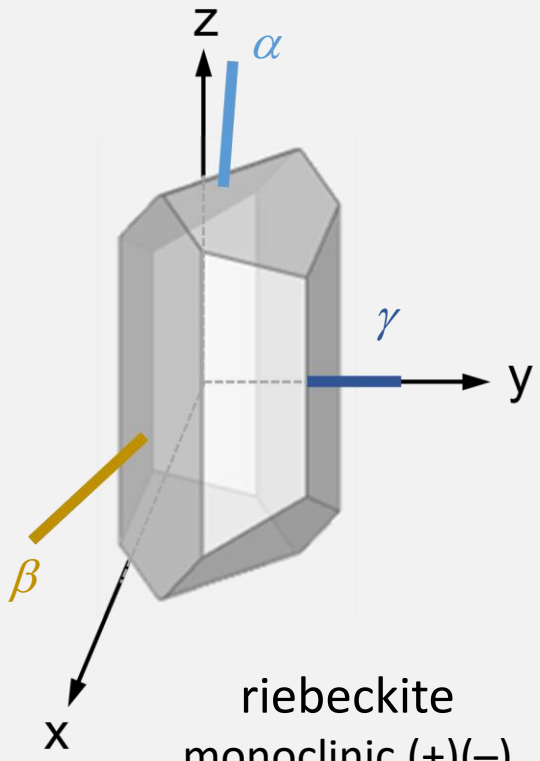
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Anisotropic minerals in PPL: Biaxial

optic axis sections

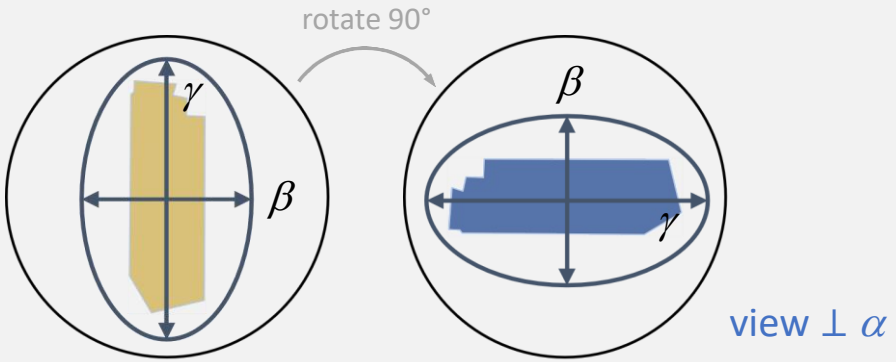


E-W polarizer

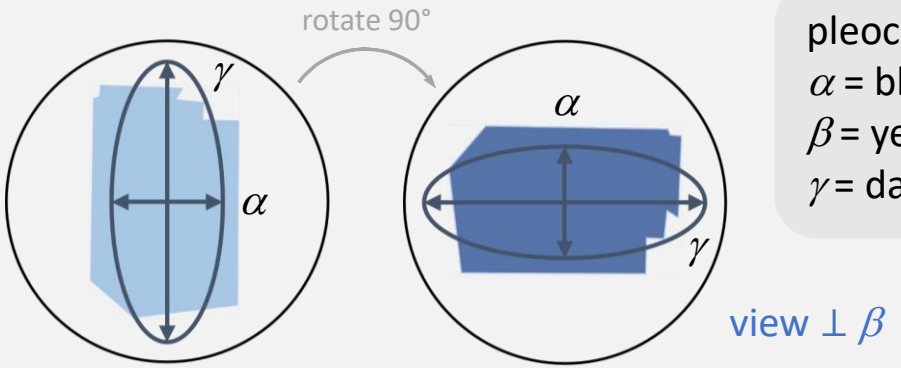


riebeckite
monoclinic (+)(–)

view ⊥ to vibration directions

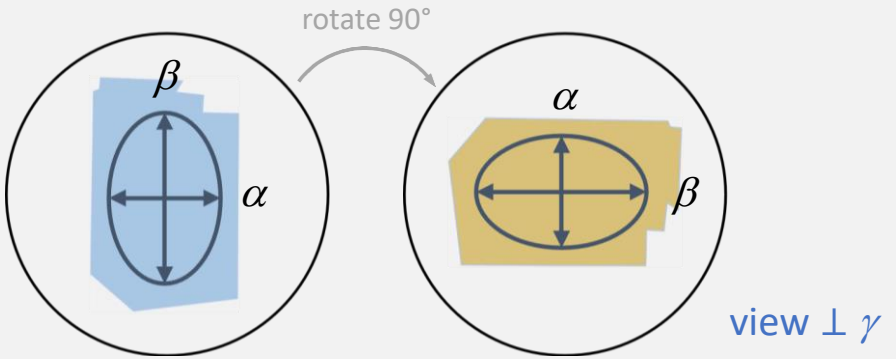


view ⊥ α



view ⊥ β

pleochroic scheme:
α = blue
β = yellow-brown
γ = dark blue



view ⊥ γ

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Optics and mineralogy

You should now be able to:

- Explain the concept of pleochroism
- Predict the observed pleochroic colour for an anisotropic mineral in a given orientation



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Optics and mineralogy

Optical properties of anisotropic minerals in crossed polars

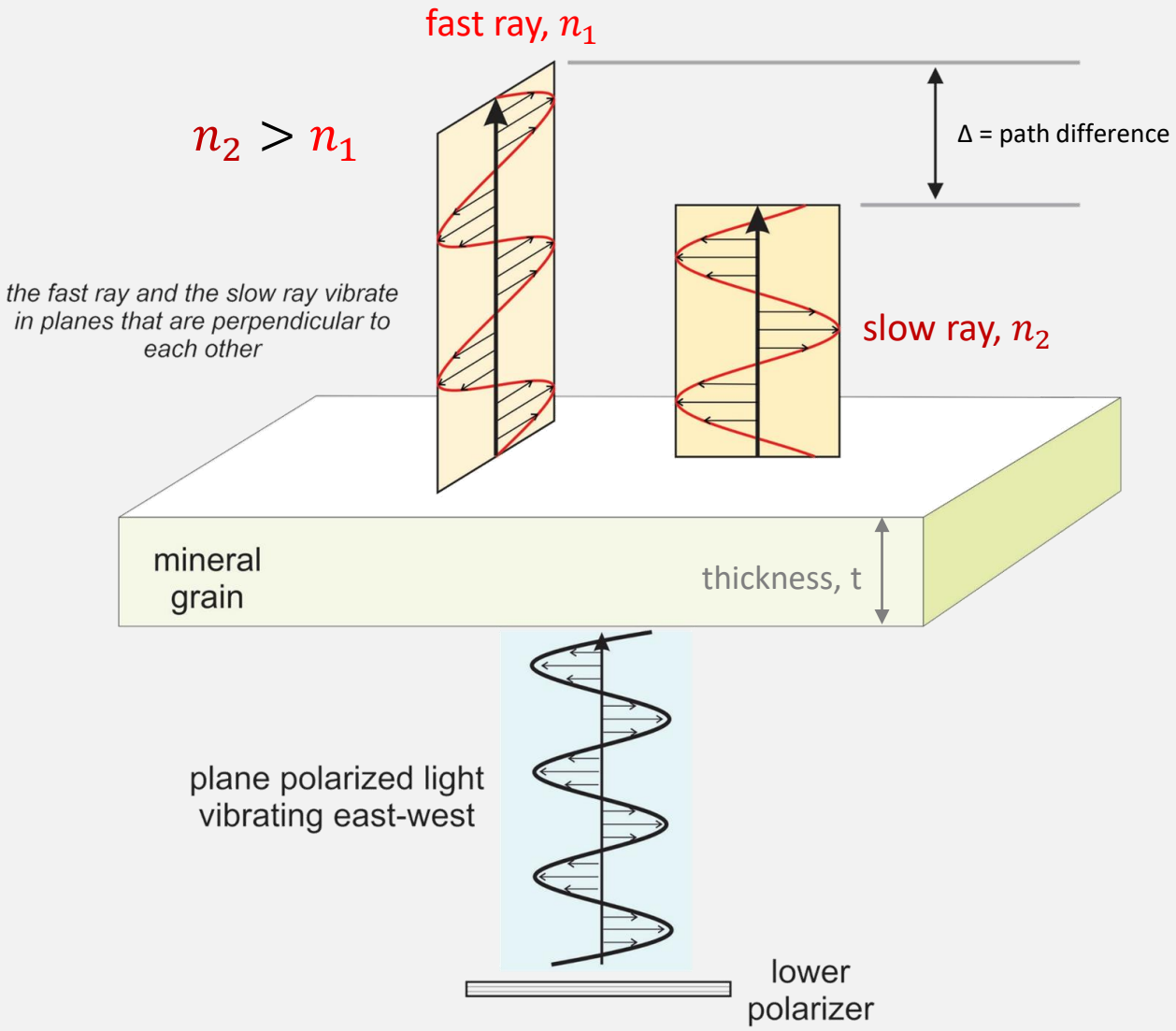
5.1 The optical path of light through anisotropic minerals

5.2 Birefringence and the Michel-Lévy chart



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The optical path of light through anisotropic minerals

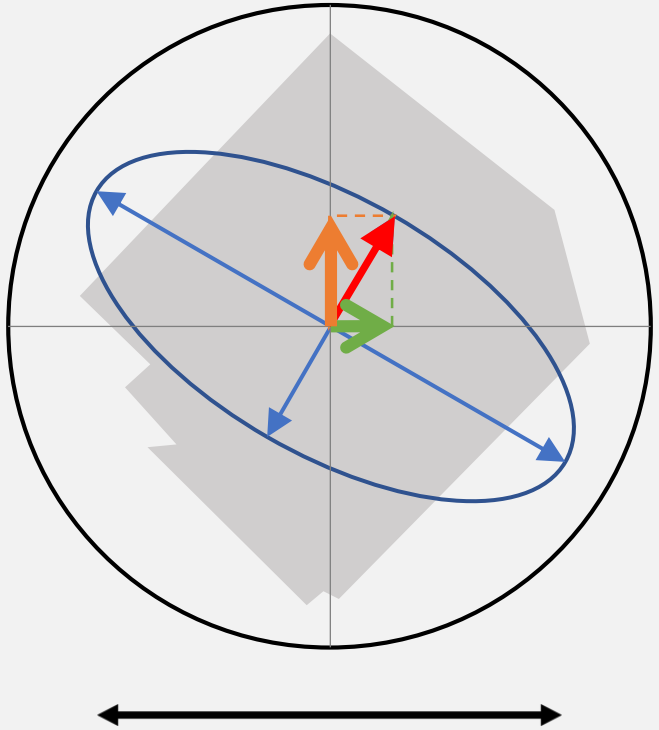


- The refractive indices are also referred to as permitted vibration directions (PVDs)
- When polarized light enters an anisotropic crystal, the light splits into two polarized rays, each travelling along one of the PVDs. This is called **double refraction**.
- The two rays travel through the mineral at different speeds, so there is an optical path difference Δ between them
- When light exits the mineral, the two rays are vibrating perpendicular to each other

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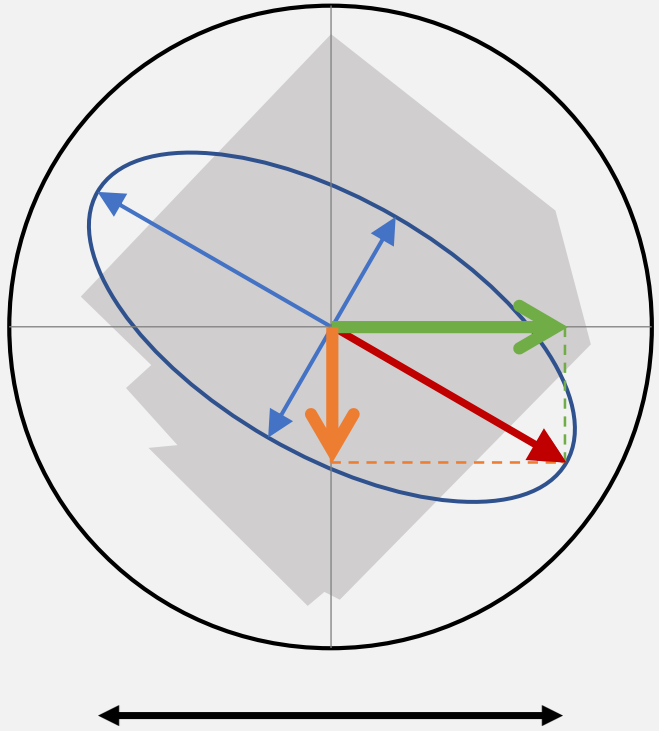
Can doubly refracted light pass through the analyser?

E-W and **N-S** components of **fast ray**



E-W polarizer

E-W and **N-S** components of **slow ray**



E-W polarizer

Each ray can be described as the sum of two vectors vibrating in the directions of the **E-W** polarizer and the **N-S** analyser



When the analyser is inserted:

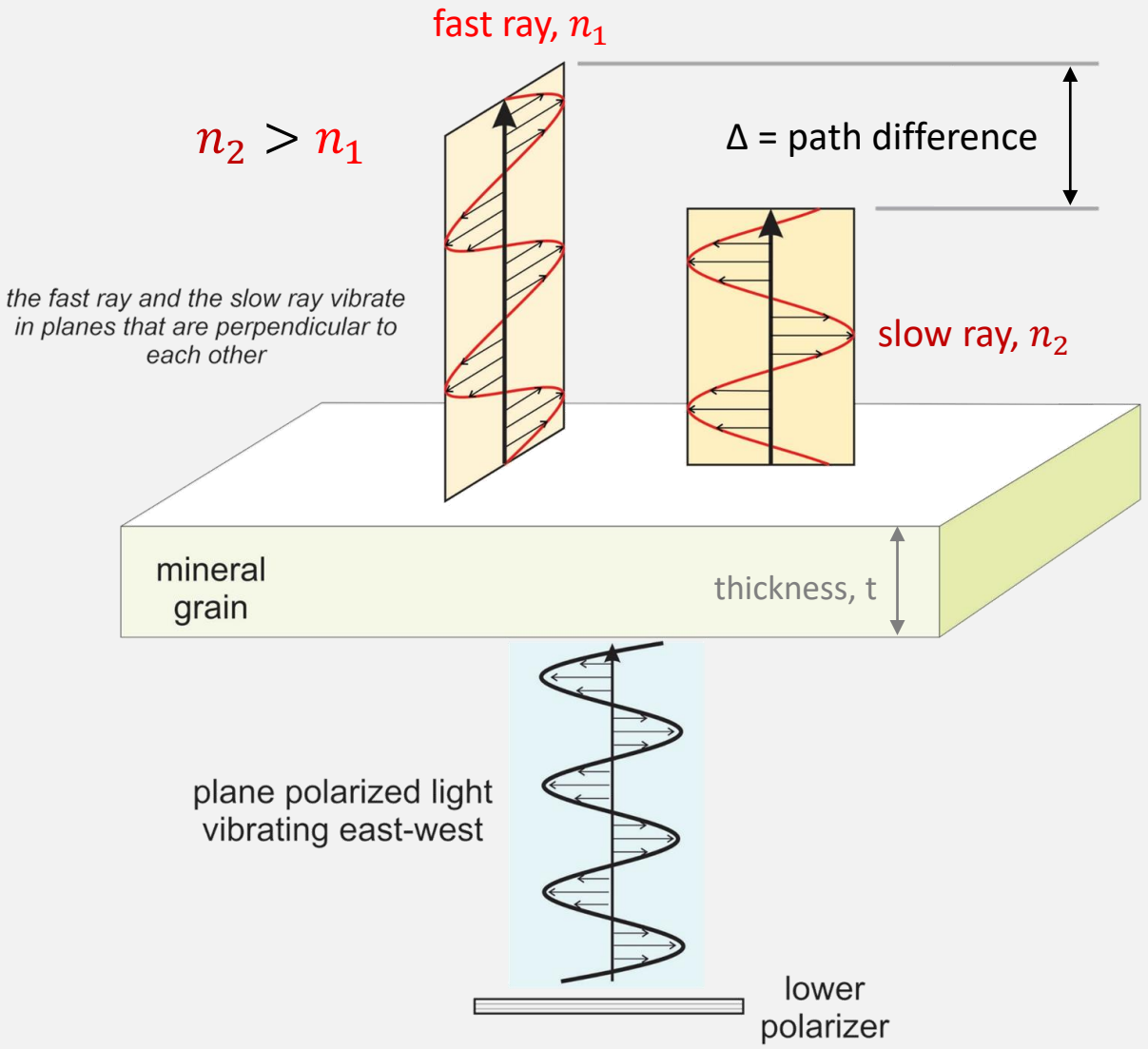
N-S components pass through

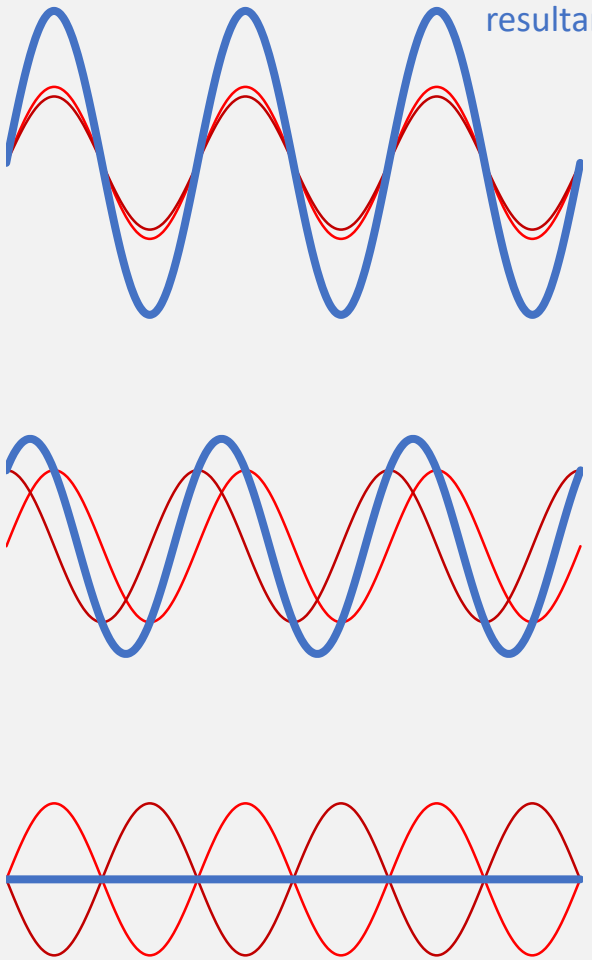
E-W components are excluded

The **N-S** components of the **fast** and **slow** rays are recombined when they pass through the analyser. The path difference between them causes interference.

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Optical path difference between fast and slow rays causes interference



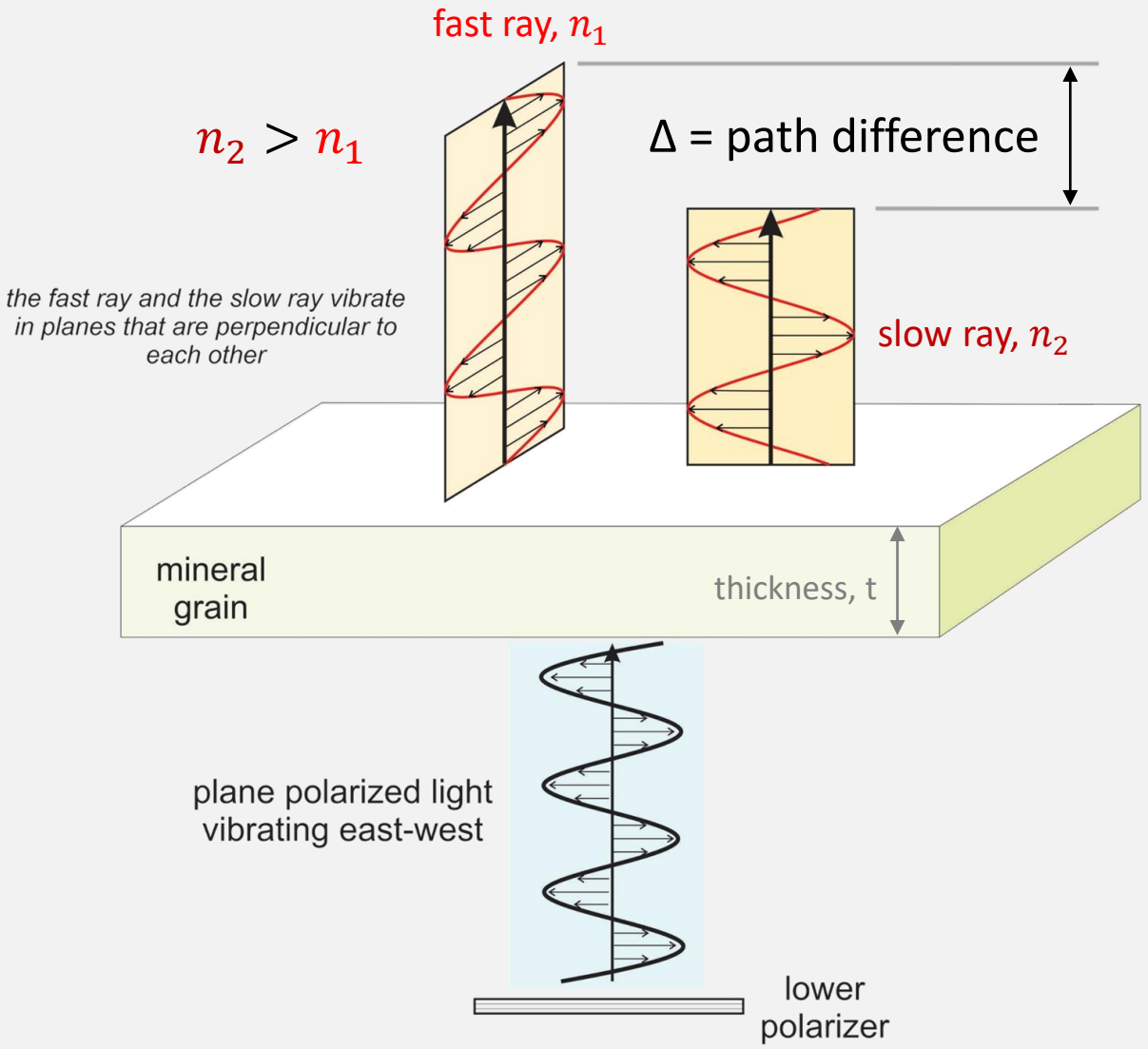


Path difference = $n\lambda$: constructive interference

Path difference = $(n+½)\lambda$: destructive interference

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Optical path difference, thickness, birefringence



$$\boldsymbol{\Delta = \delta \times t}$$

For $n_2 > n_1$: $\Delta = t(n_2 - n_1)$

For $n_2 < n_1$: $\Delta = t(n_1 - n_2)$

birefringence $\boldsymbol{\delta = |n_2 - n_1|}$

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Optics and mineralogy

Optical properties of anisotropic minerals in crossed polars

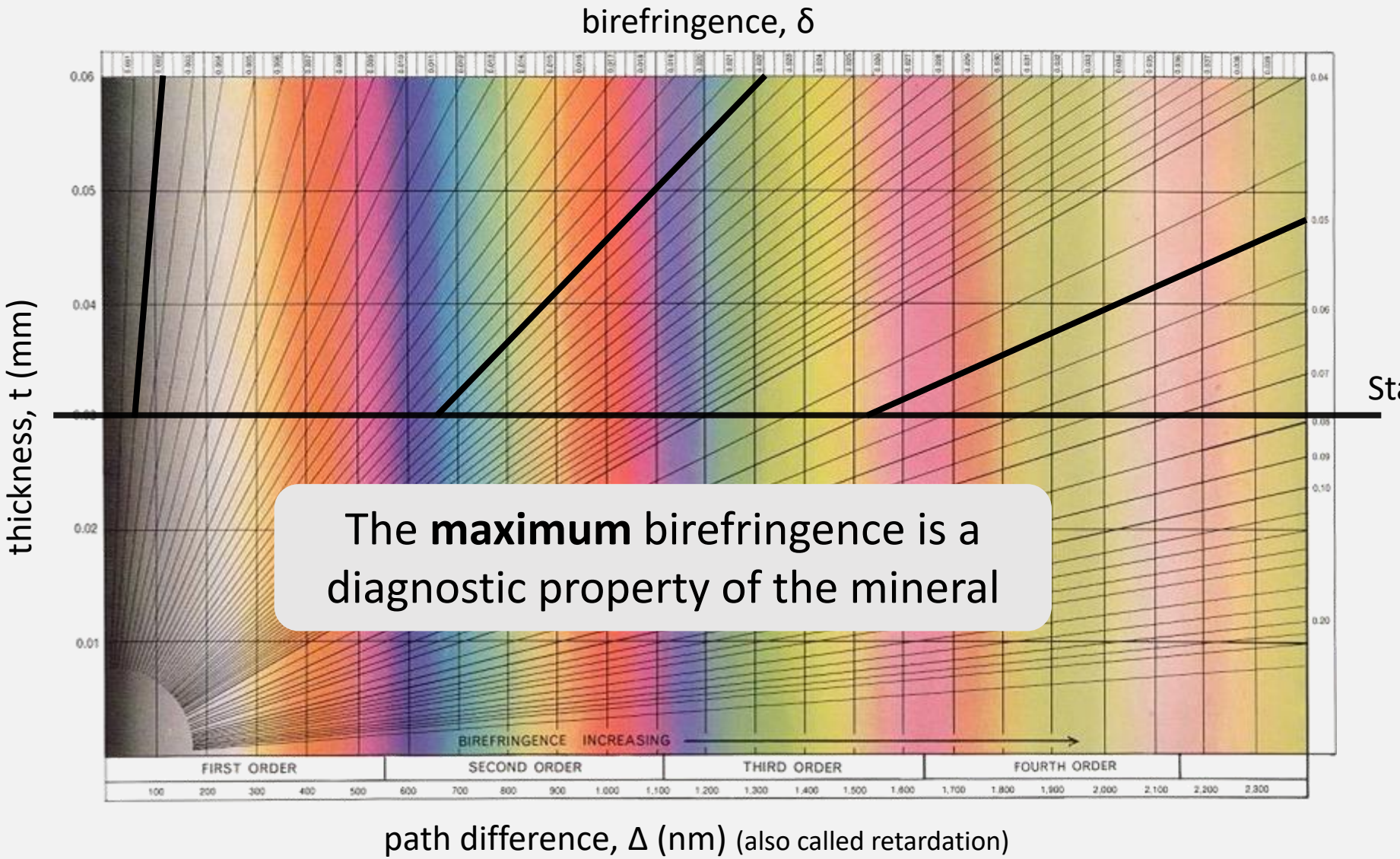
5.1 The optical path of light through anisotropic minerals

5.2 Birefringence and the Michel-Lévy chart



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Birefringence and interference colours: the Michel-Lévy chart



The **maximum** birefringence is a diagnostic property of the mineral

$$\Delta = \delta \times t$$

Standard thin section: t = 30 µm

Uniaxial minerals:

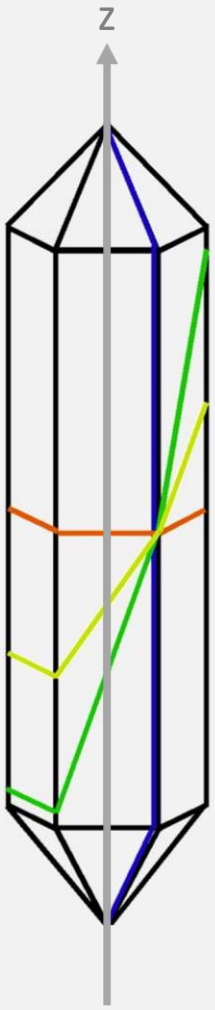
maximum $\delta = |\varepsilon - \omega|$

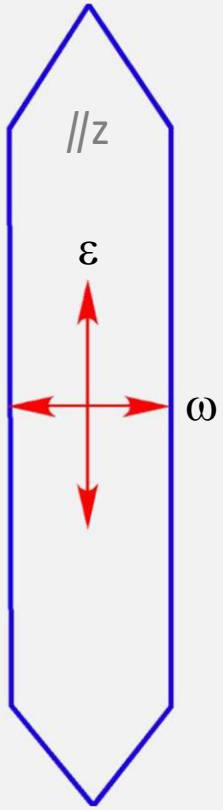
Biaxial minerals:

maximum $\delta = |\gamma - \alpha|$

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The observed interference colour depends on how the crystal is sectioned

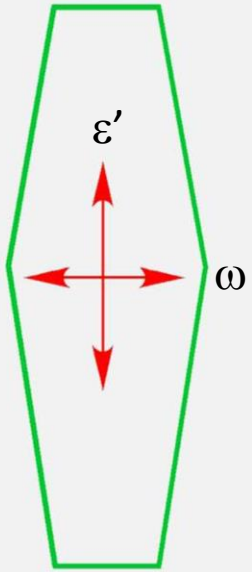




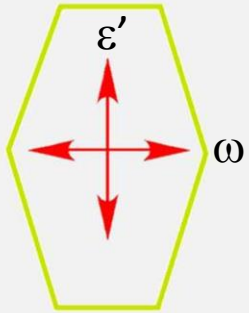
δ = 0.09
yellow-white

Quartz: Trigonal

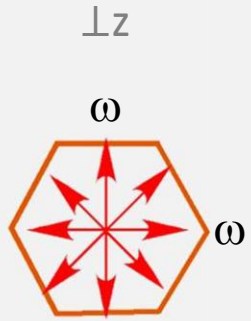
Refractive indices: ε = 1.553
ω = 1.544



δ = 0.05
pale grey



δ = 0.02
dark grey

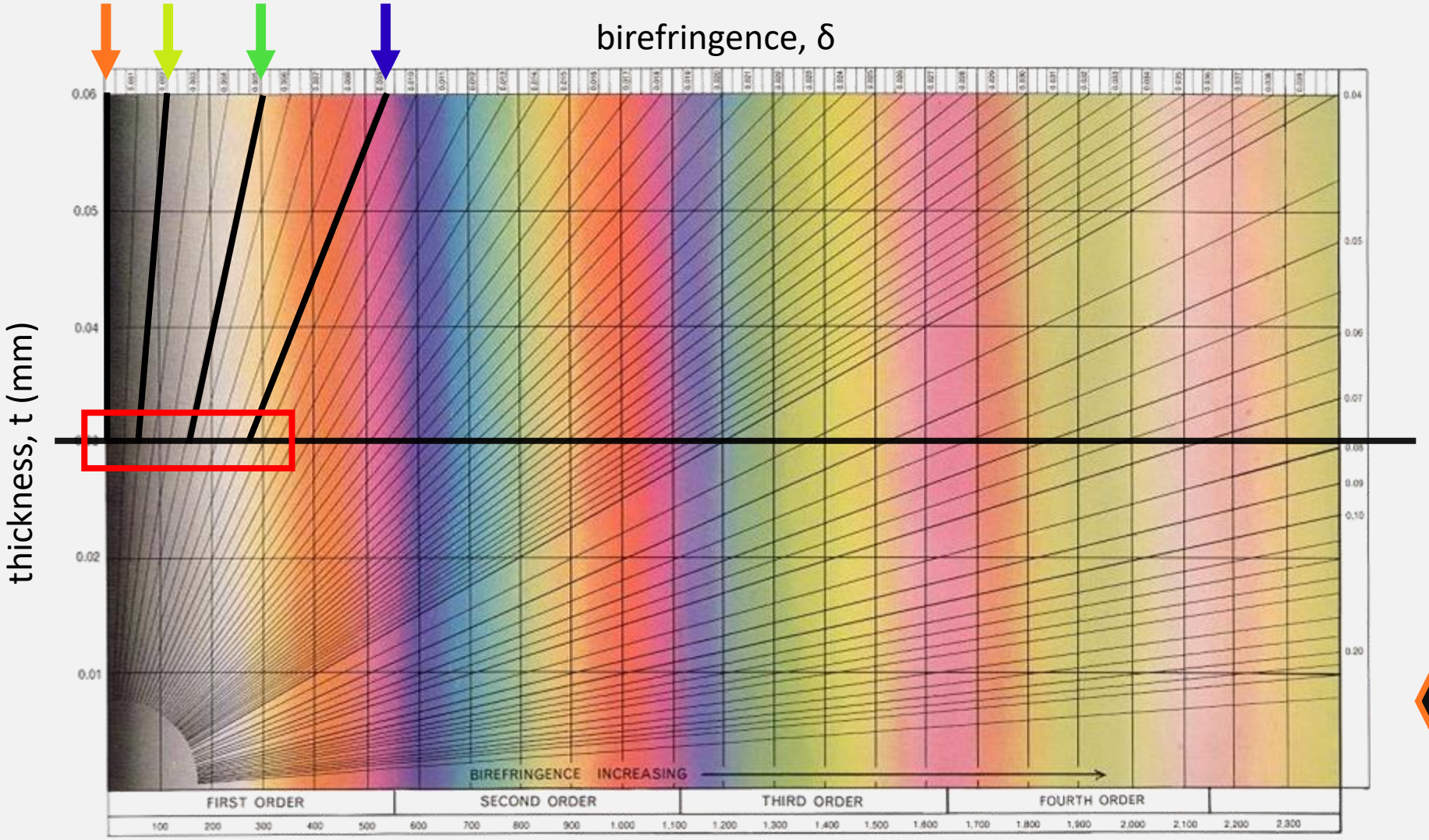


δ = 0
isotropic

Maximum birefringence of quartz is 0.09

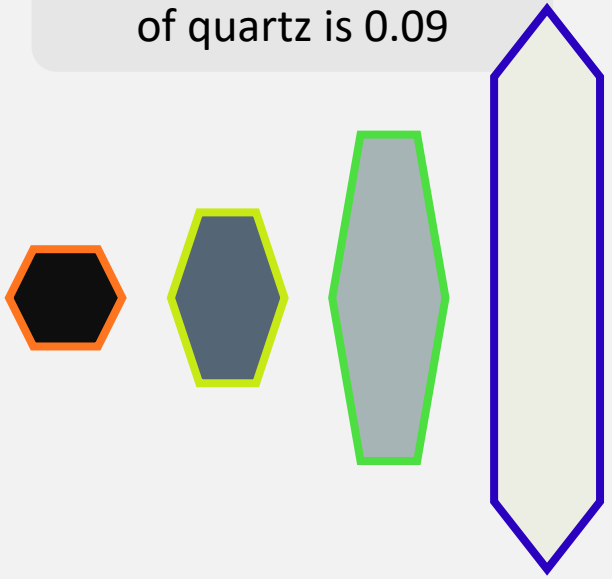
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The observed interference colour depends on how the crystal is sectioned



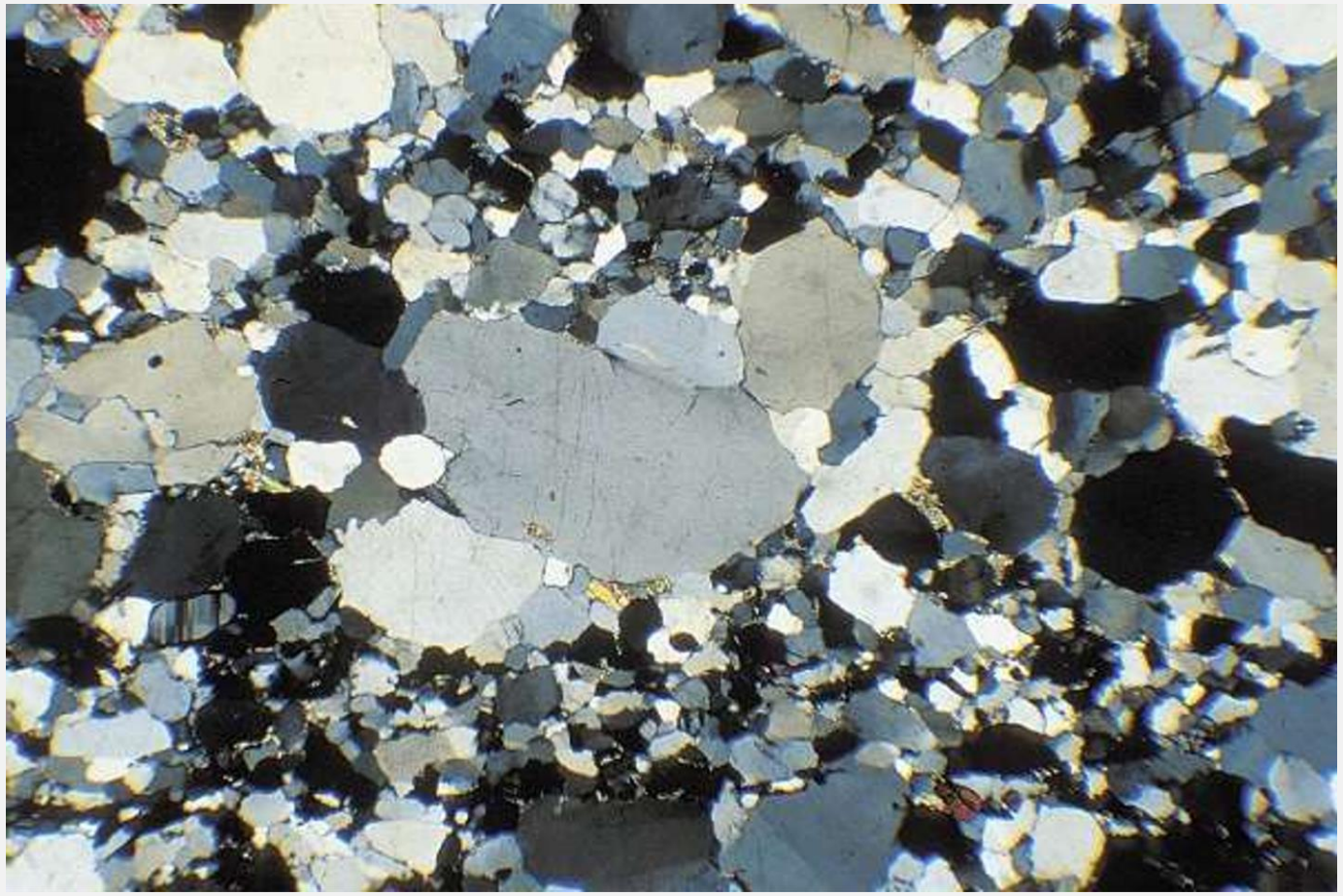
$$\Delta = \delta \times t$$

Maximum birefringence of quartz is 0.09



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The observed interference colour depends on how the crystal is sectioned

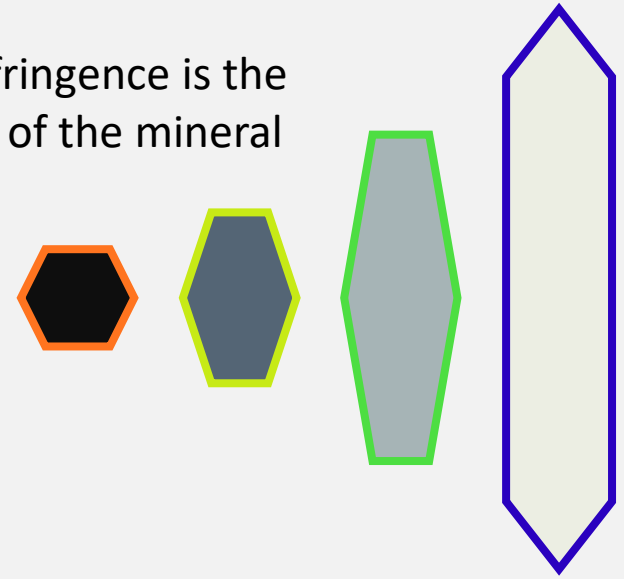


Quartz in thin section

Notice the wide range of interference colours

- **δ = 0**: extinct, the isotropic optic axis section
- **δ = 0.09**: white, the maximum birefringence
- **0 < δ < 0.09**: all intermediate interference colours seen in the thin section

The **maximum** birefringence is the diagnostic property of the mineral



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Optics and mineralogy

You should now be able to:

- Explain the concept of optical interference
- Use the Michel-Lévy chart to determine the interference colour associated with a given value of birefringence, and vice versa
- State that the maximum birefringence is a diagnostic property of a mineral
- Explain why, in a single thin section, different slices through mineral might show different interference colours



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Optics and mineralogy

Optical properties of anisotropic minerals in crossed polars

6.1 Extinction in uniaxial minerals

6.2 Extinction in biaxial minerals

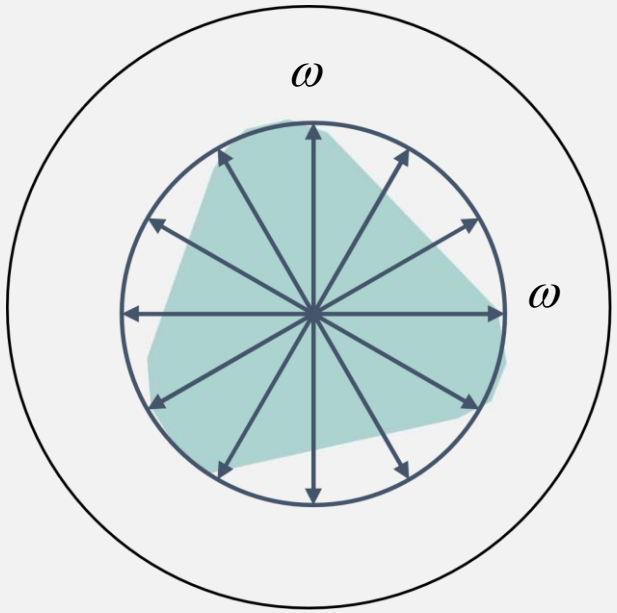
6.3 Types of extinction and relationship to symmetry



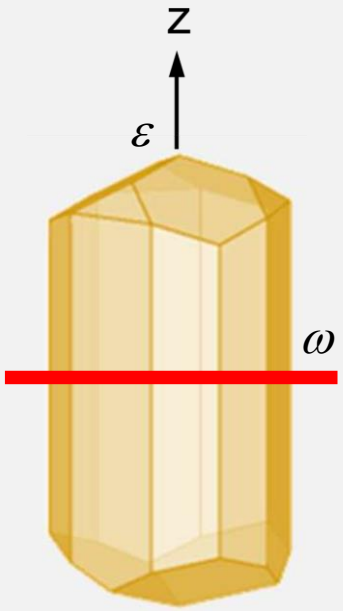
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Anisotropic minerals in PPL: Uniaxial

optic axis (basal) section



E-W polarizer

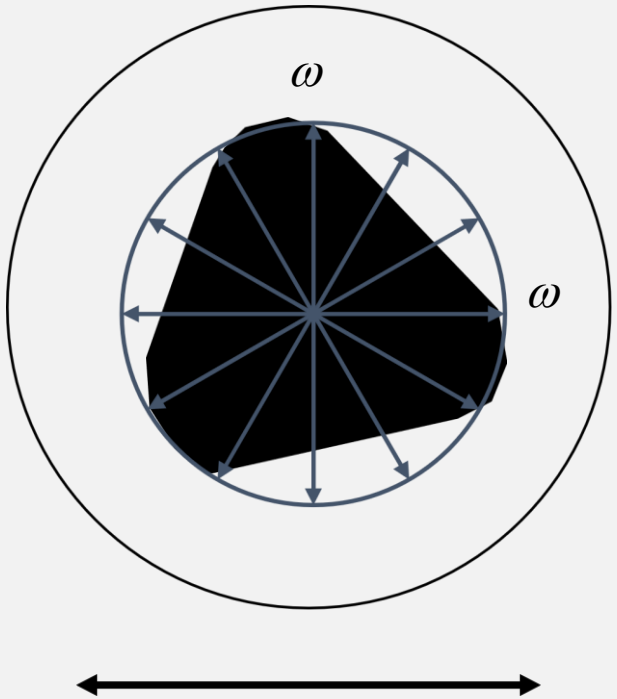


tourmaline
trigonal (–)

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Anisotropic minerals in XP: Uniaxial

optic axis (basal) section

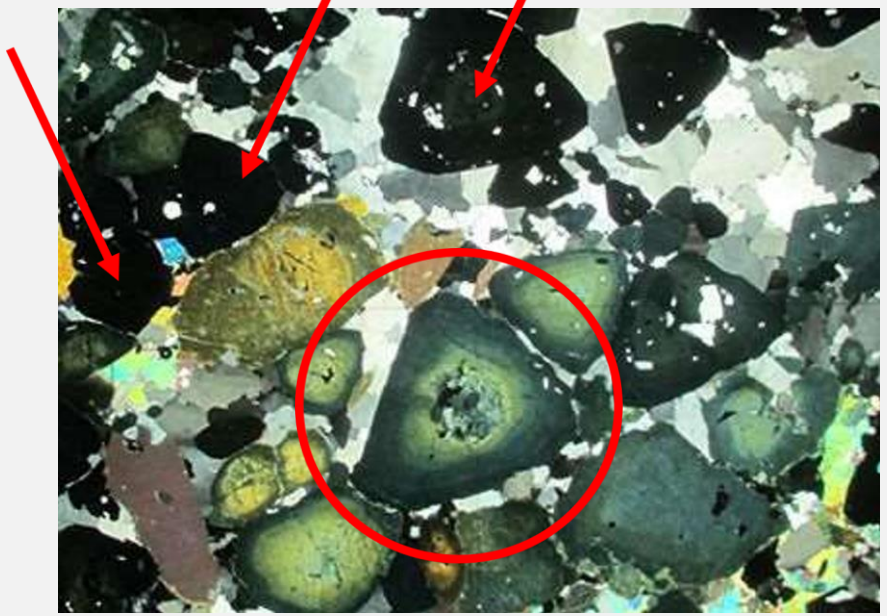


E-W polarizer

N-S analyser

tourmaline
trigonal (–)

crossed polars

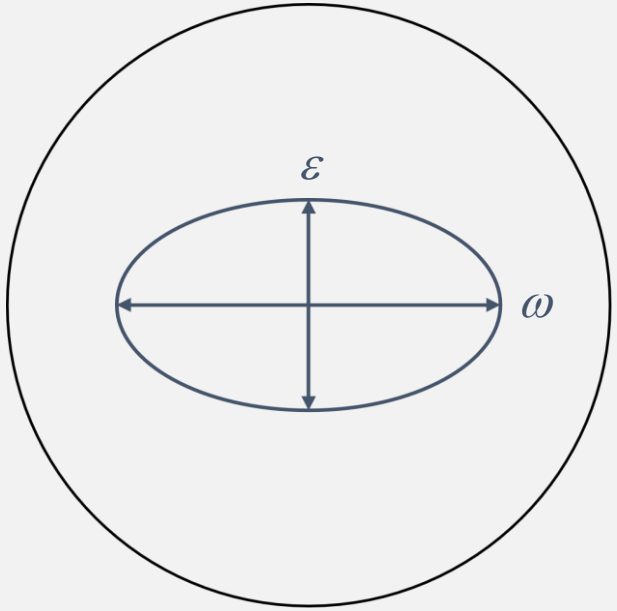


extinction

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Anisotropic minerals in XP: Uniaxial

prismatic section

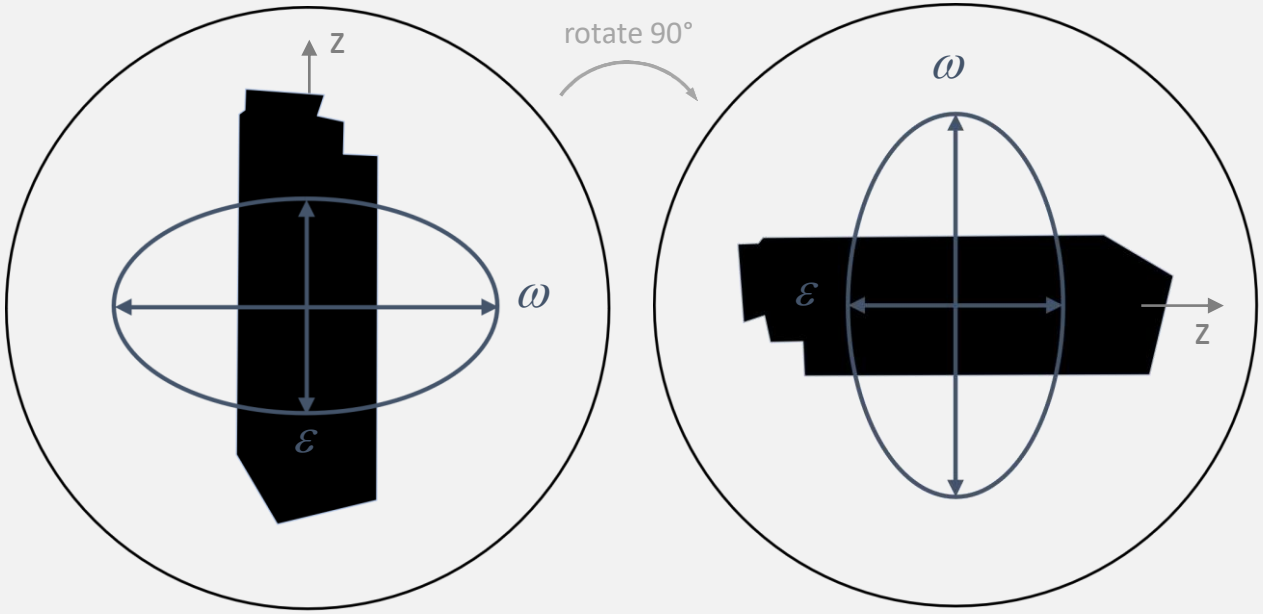


E-W polarizer

N-S analyser

tourmaline
trigonal (–)

vibration directions aligned with polarizers



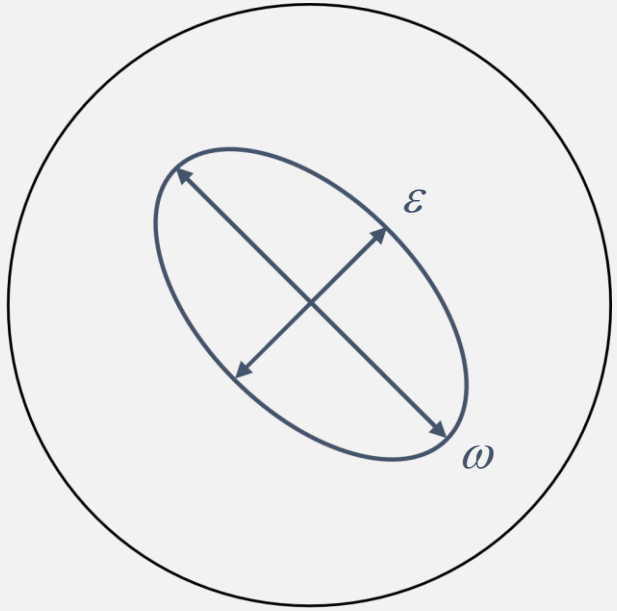
E-W polarizer

extinction

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Anisotropic minerals in XP: Uniaxial

prismatic section

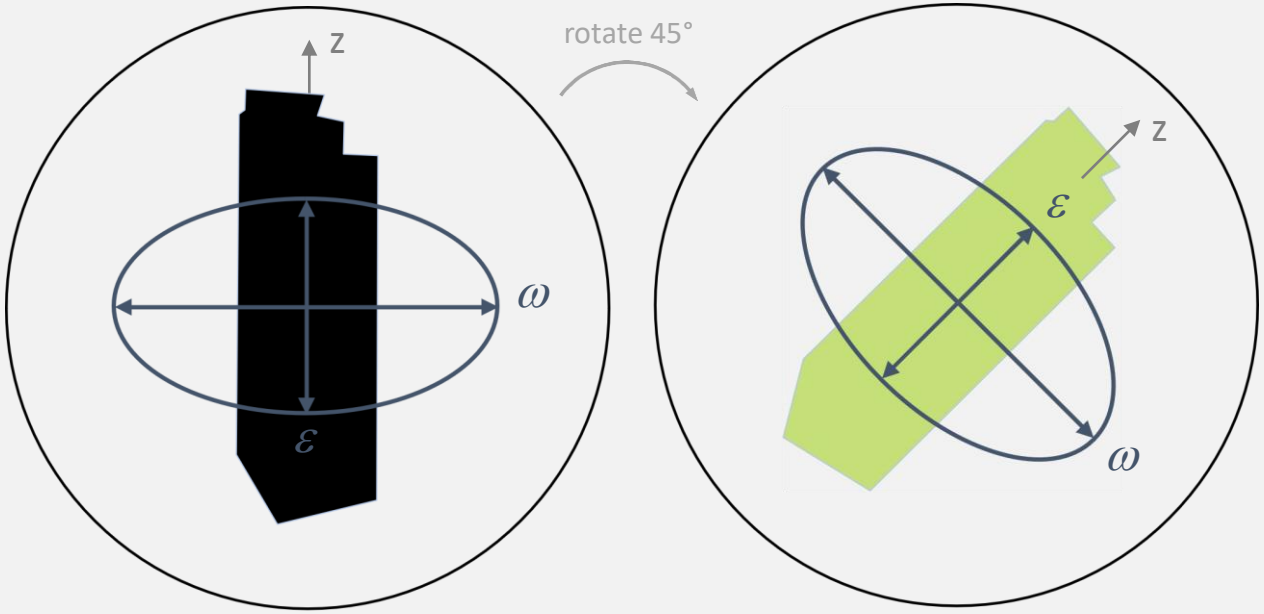


E-W polarizer

N-S analyser

tourmaline
trigonal (–)

vibration directions at 45° to polarizers



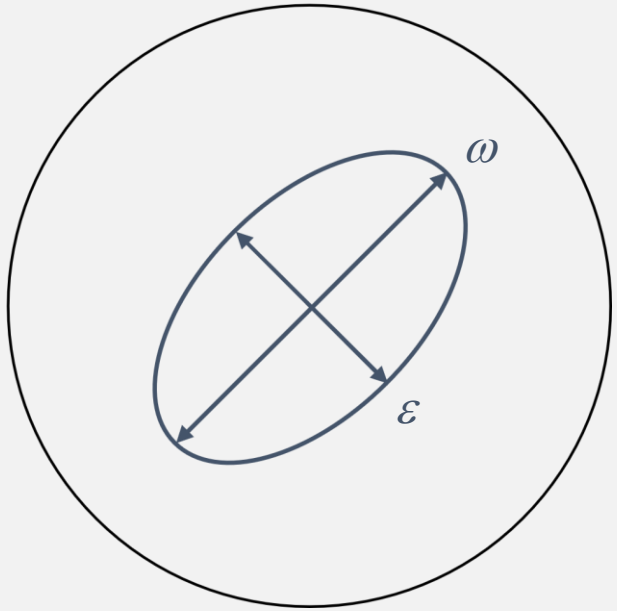
E-W polarizer

interference colour

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Anisotropic minerals in XP: Uniaxial

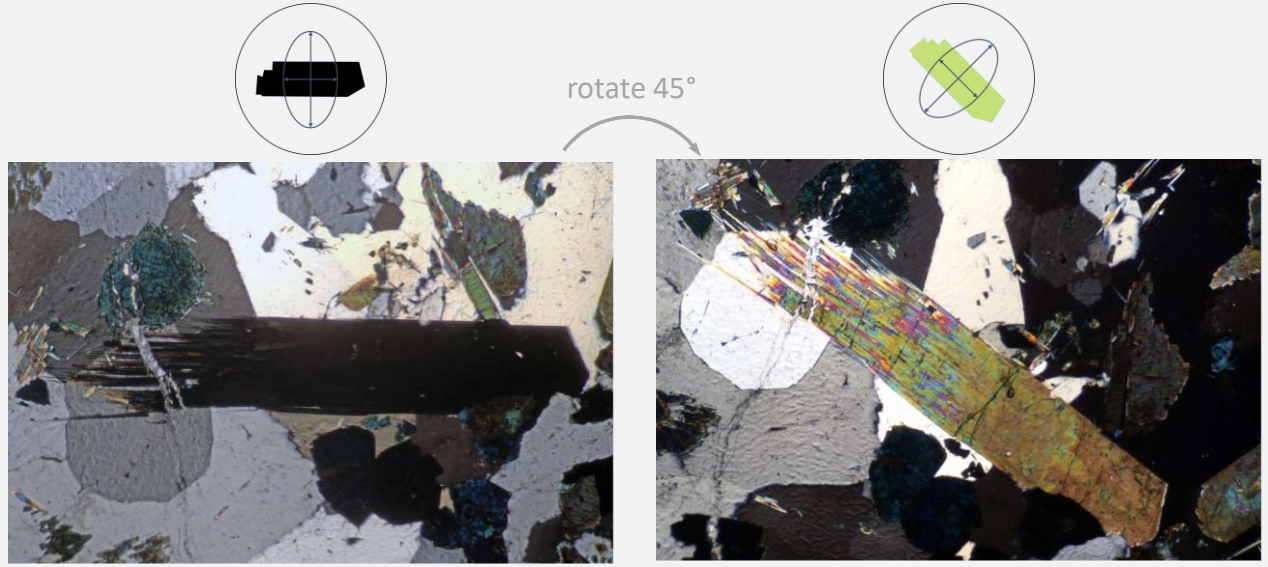
prismatic section



E-W polarizer

N-S analyser

tourmaline
trigonal (–)

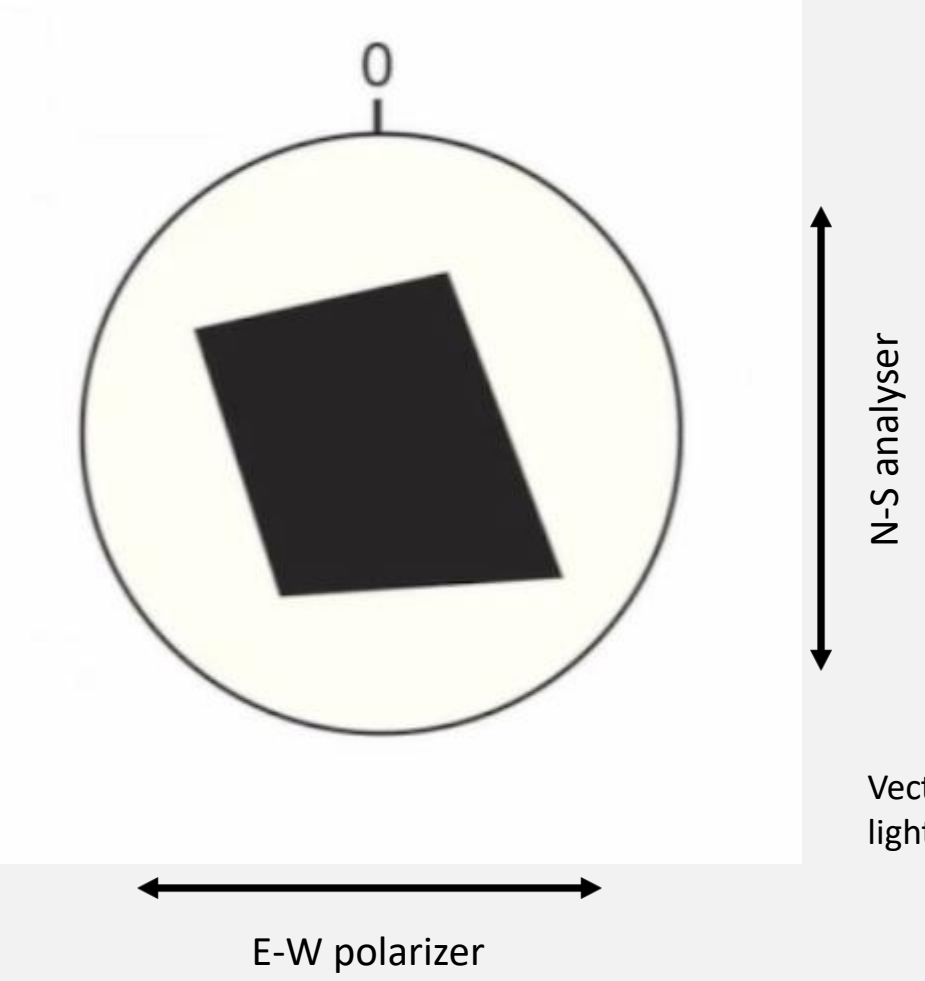


straight extinction

interference colour

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Key observation: in any anisotropic section through a mineral, extinction occurs every 90°



- Light intensity varies with stage rotation
- When the refractive indices and PVDs are parallel to the polarizer and analyser, no light gets through. The mineral is in **extinction**
- Maximum illumination occurs at 45° to extinction
- Extinction occurs every 90°. This is true for all anisotropic slices through all anisotropic minerals, so this observation is not useful in a thin section description.

Vectors show the components of light in the **E-W** and **N-S** directions

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Optics and mineralogy

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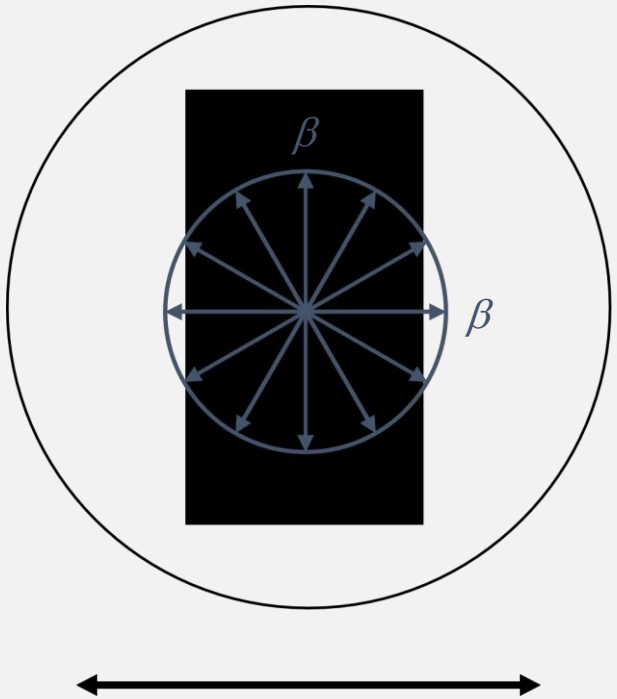
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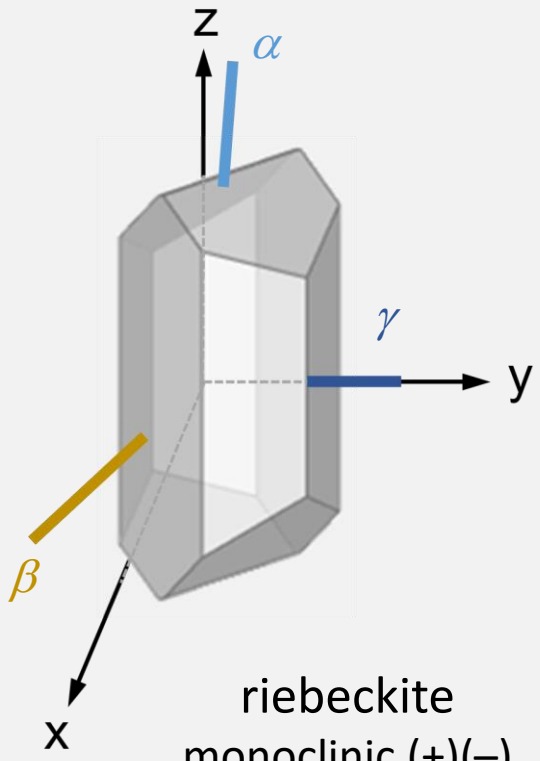
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Anisotropic minerals in XP: Biaxial

optic axis sections

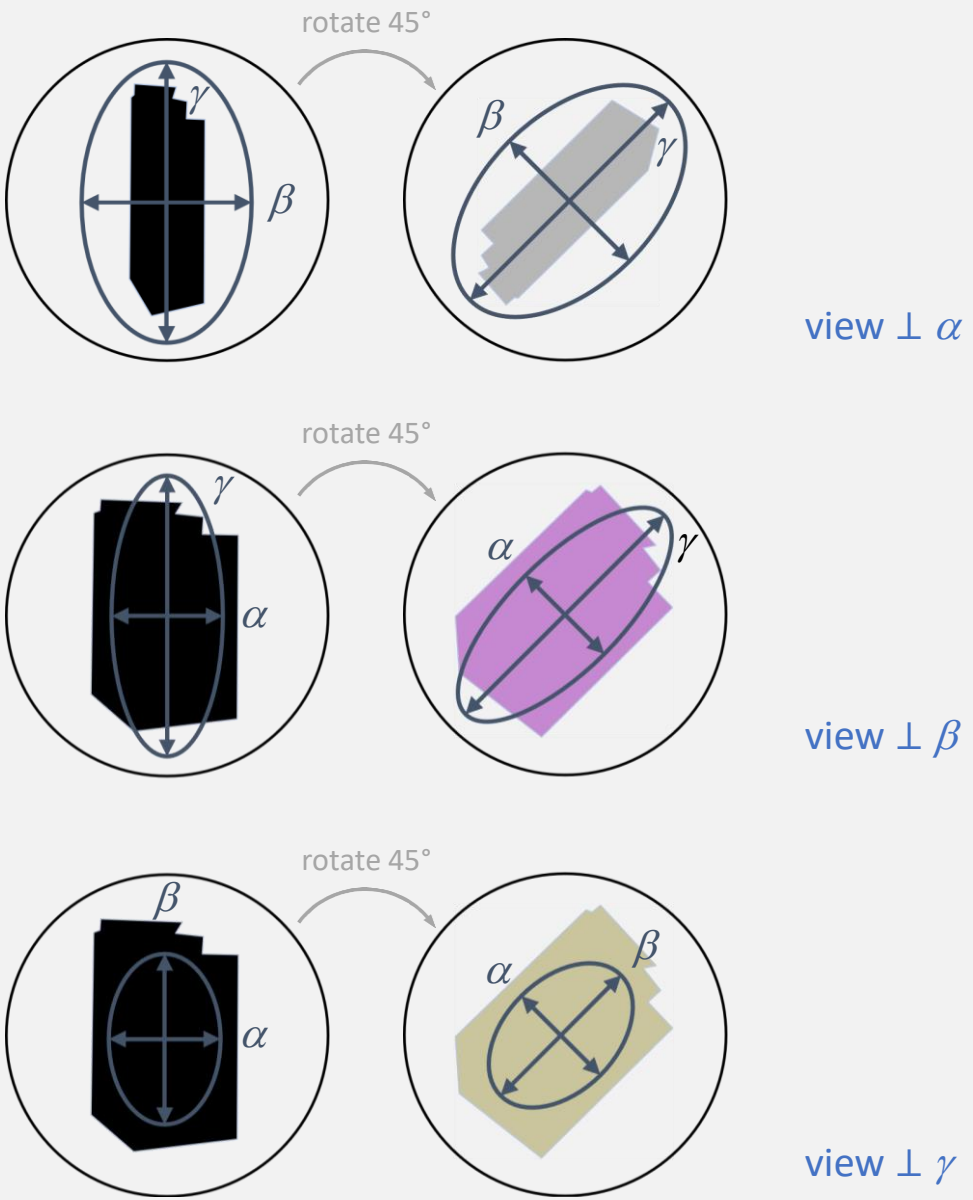


E-W polarizer



riebeckite
monoclinic (+)(–)

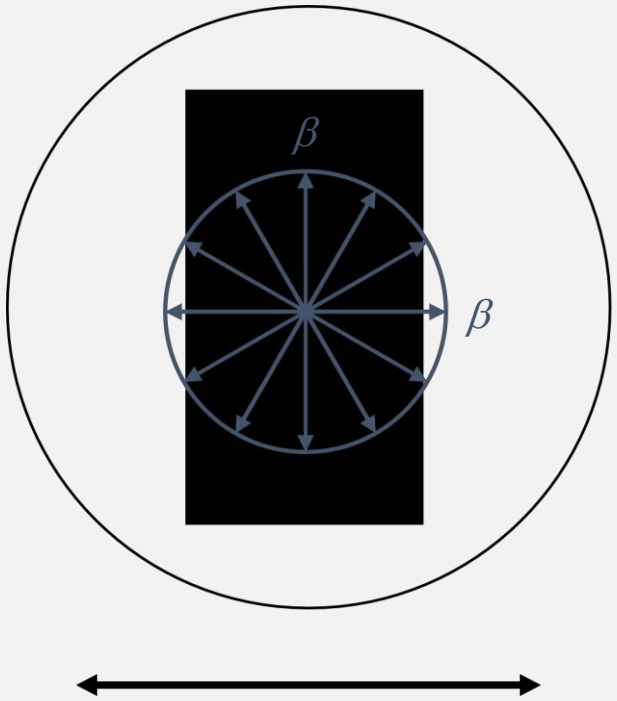
view ⊥ to vibration directions



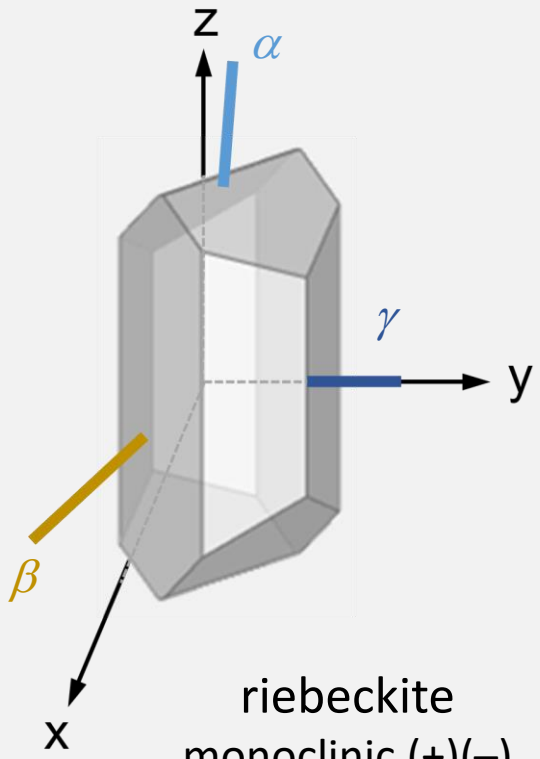
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Anisotropic minerals in XP: Biaxial

optic axis sections

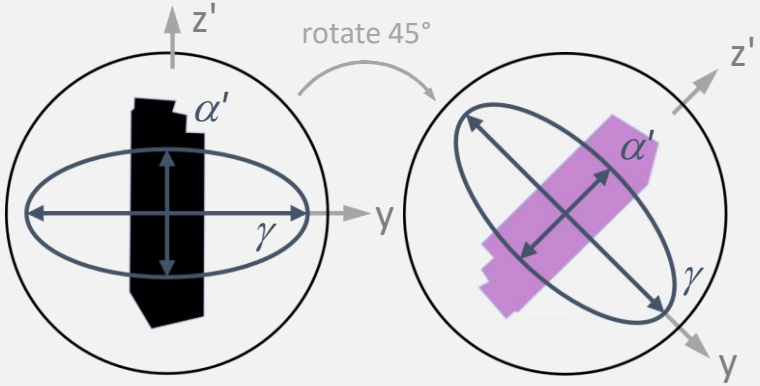


E-W polarizer

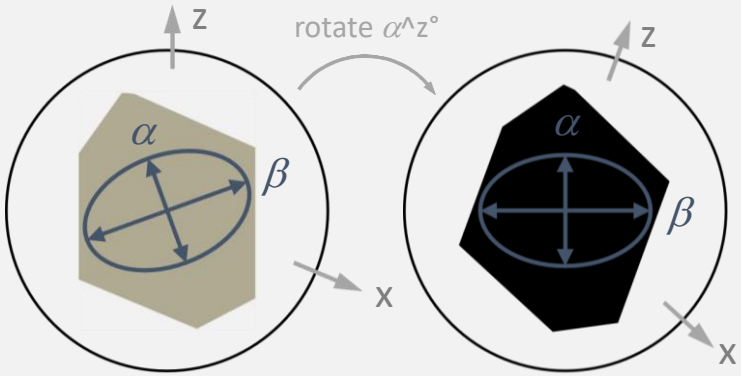


riebeckite
monoclinic (+)(–)

view ⊥ to crystallographic axes

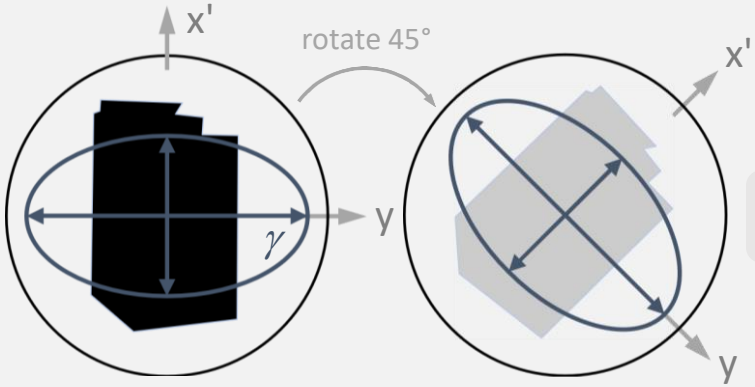


view ⊥ x



inclined extinction

view ⊥ y



straight extinction

view ⊥ z

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Optics and mineralogy

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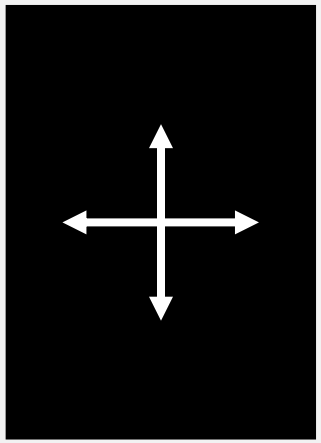


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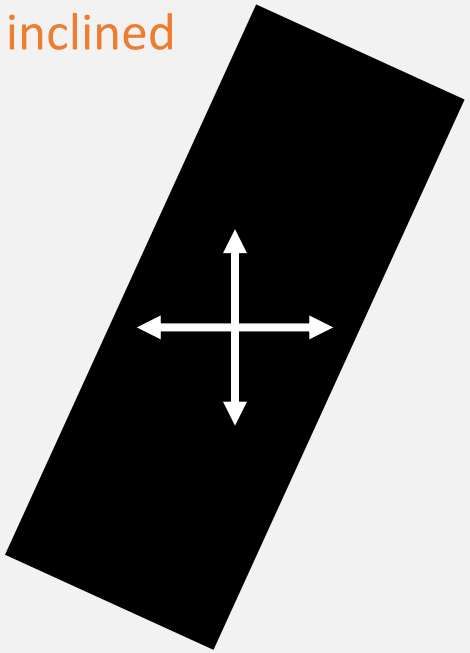
How to describe extinction

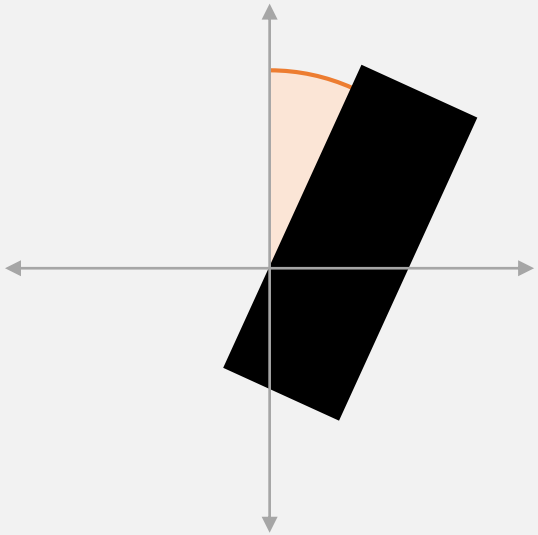
Extinction described with respect to crystal edges:

straight



inclined





The extinction angle is a diagnostic feature of particular minerals and mineral groups

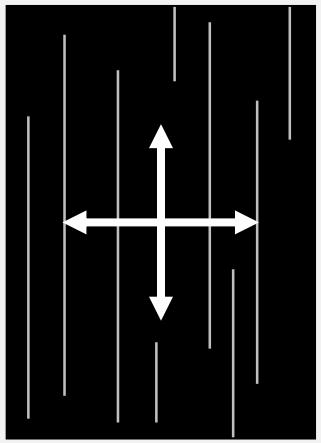
Extinction is the position at which a mineral appears black under crossed polars

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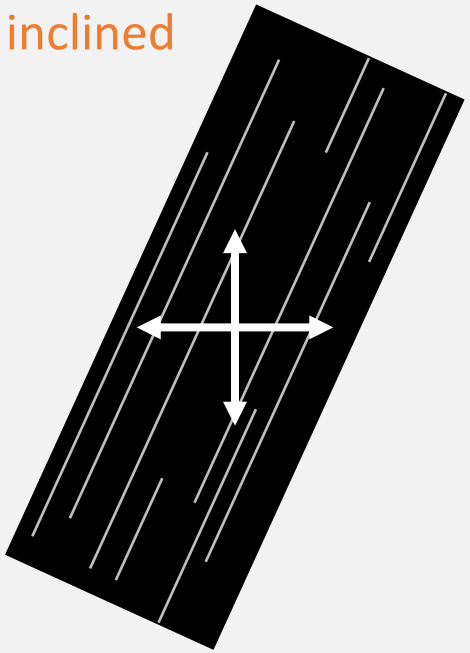
How to describe extinction

Extinction described with respect to cleavage traces:

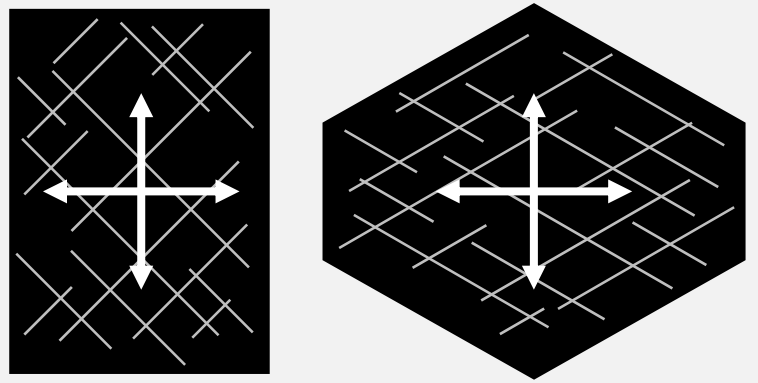
straight



inclined



symmetrical



Extinction is the position at which a mineral appears black under crossed polars

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Crystal system	Indicatrix classification	Defining RIs	Types of extinction in XPL
Cubic	isotropic	n	always in extinction
Trigonal (rhombohedral)			
Tetragonal	uniaxial	ε and ω	straight extinction parallel to the z axis
Hexagonal			
Orthorhombic			straight extinction parallel to the x, y and z axes
Monoclinic	biaxial	α, β and γ	straight extinction in sections containing the y axis; inclined extinction in all other sections
Triclinic			no sections constrained to have straight extinction

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Optics and mineralogy

You should now be able to:

- Explain the concept of extinction
- Define the optical conditions under which a mineral will be in extinction in thin section
- Identify straight, inclined and symmetrical extinction in a thin section
- Predict the orientation at which an anisotropic mineral will be in extinction using information about the crystal system and indicatrix



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Summary so far

Mineral properties visible in plane-polarized light (PPL)	Mineral properties visible under crossed polars (XP)
Crystal shape and habit	Interference colour (birefringence)
Cleavage	Extinction
Colour and pleochroism	Twinning
Relief	Zoning

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